

Brine Dispersion Modelling Case Study

Universal Brine Dispersion Solver (UBDS): Near-, Medium- and Far-Field Assessment of the Bahia Azul Seawater Desalination & Brine Outfall

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Solver: UBDS v1.0.0 | Document: case.docx

Executive Summary

This study presents a holistic near-, medium- and far-field assessment of the brine discharge from the proposed Bahia Azul seawater desalination and brine outfall, performed entirely with the Universal Brine Dispersion Solver (UBDS) - a novel, open, physics-based solver developed by the author. UBDS unifies, in a single dynamically-coupled and mass-conserving framework, the near-field integral jet/plume modelling and the far-field hydrodynamic transport modelling that conventionally require two separate, proprietary codes. Its sign-agnostic buoyancy treatment and dense-water gravity-current closure make it universal across negatively buoyant (brine), positively buoyant (thermal) and neutral discharges.

For the proposed inclined two-port diffuser, the near-field initial dilution of the reverse-osmosis (RO) concentrate at first seabed contact ranges between $\times 17$ and $\times 30$ across the flow and tidal range, corresponding to seabed salinities of 37.6-38.4 psu against a background of 36.5 psu.

The medium- and far-field assessment confirms that, at the maximum discharge of 1200 m³/hr, the salinity increment of 0.5 psu above background is confined to an area of 1.062 km², with the 38.5 psu threshold not exceeded beyond 0 m from the diffuser - well within the 100 m regulatory mixing zone. All trace impurities remain below their Environmental Quality Standards after dilution.

The solver was validated against canonical self-similar jet and plume dilution laws, published inclined dense-jet experimental data, reference seawater/brine densities, and a synthetic ADCP current-meter dataset; all benchmarks are met. Data sources are recorded in the Validation section.

1 Introduction and the UBDS Solver

Brine discharges from desalination plants and solution-mined storage caverns are negatively buoyant and sink towards the seabed, where elevated salinity can affect benthic ecology. Their assessment requires resolving both the small-scale turbulent jet/plume mixing in the immediate vicinity of the diffuser (the near field) and the tidally-driven transport and dilution over the wider water body (the medium and far field). Conventionally these two regimes are handled by separate, proprietary software, with manual transfer of results between them.

1.1 Novelty and gaps bridged

- Unified near-to-far field coupling in a single open framework, removing the manual hand-off and the associated mass-conservation errors between separate codes.
- Sign-agnostic buoyancy: the same governing equations describe dense (brine), buoyant (thermal/freshwater) and neutral discharges - a genuinely universal solver.
- A dense-water gravity-current closure in the sigma-layer transport engine that reproduces the strong bottom-trapping of brine that 2-D depth-averaged models cannot capture and that 3-D models obtain only at large computational cost.
- Built-in regulatory-metric engine: mixing-zone radius, salinity-increment areas, diffusion distances and EQS compliance computed automatically from the solution.
- Fully open and reproducible: physically-standard, documented entrainment and turbulence closures, with all inputs and outputs exposed as data.

1.2 Governing equations

Near field - steady Eulerian flux-integral equations (top-hat) integrated along the jet arclength s , with combined shear + cross-flow entrainment and a sign-agnostic buoyancy body force:

$$d(\rho_e Q)/ds = \rho_a E; \quad d(\rho_e Q V)/ds = \rho_a E V_a + (0, 0, (\rho_a - \rho_e) g \pi b^2)$$

$$d(\rho_e Q S)/ds = \rho_a E S_a; \quad E = 2 \pi b \alpha_s |V - V_a| + 2 b \alpha_f |V_{a_perp}|$$

Far field - depth-averaged shallow-water hydrodynamics (mass + momentum with Manning friction, Coriolis and turbulent mixing) driving a sigma-layer advection-dispersion equation augmented with the buoyancy-scaled inter-layer gravity-current flux:

$$d(\eta)/dt + d(Hu)/dx + d(Hv)/dy = 0$$

$$d(hs)/dt + d(uhs)/dx + d(vhs)/dy = d/dx(hDx ds/dx) + d/dy(hDy ds/dy) + \text{sources}$$

2 Case Study Site and Input Data

Site: Bahia Azul Seawater Desalination & Brine Outfall (Bahia Azul (fictional), open Mediterranean-type coast). A mid-size seawater reverse-osmosis desalination plant (product capacity 60,000 m³/d) discharging hypersaline concentrate through an offshore inclined multiport diffuser, with provision for later receipt of saturated leaching brine from an adjacent salt-cavern energy-storage scheme. Latitude 36.5 N.

2.1 Discharge and ambient inputs

Product (freshwater) capacity	60,000 m ³ /d
Recovery ratio	0.45
Brine (concentrate) flow - full	1200 m ³ /hr (28,800 m ³ /d)
RO brine salinity / excess temp	68.0 psu / +0.3 C
Cavern brine salinity / excess temp	250.0 psu / +2.0 C
Background salinity / temperature	36.5 psu / 18.0 C
Outfall depth / offshore distance	18.0 m / 850.0 m
Diffuser	2 x 6" ports @ 20.0 m, 60.0 deg, 0.8 m above bed
Mixing-zone radius / threshold	100.0 m / 38.5 psu

2.2 Flow scenarios

Table. Hydrodynamic and transport model parameters used in the simulations.

Parameter	Value
Minimum time step	0.4 s
CFL number	0.45
Manning coefficient n	0.025 (0.022-0.04)
Horizontal eddy viscosity	2.0 m ² /s
Smagorinsky constant	0.2
Horizontal dispersion	1.0 m ² /s
Vertical diffusivity	0.0005 m ² /s
Gravity-current coefficient	12.0
Background salinity	36.5 psu
Background temperature	18.0 C
RO brine salinity	68.0 psu
Cavern brine salinity	250.0 psu
Outfall depth	18.0 m

3 Model Geometry, Mesh and Setup

Two nested structured-grid domains were configured: a high-resolution medium-field domain (2500 x 2500 m at 25 m, 9 sigma layers) and a regional far-field domain (7200 x 7200 m at 60 m, 4 layers). The bathymetry deepens offshore and includes a southern headland whose lee generates a tidal eddy.

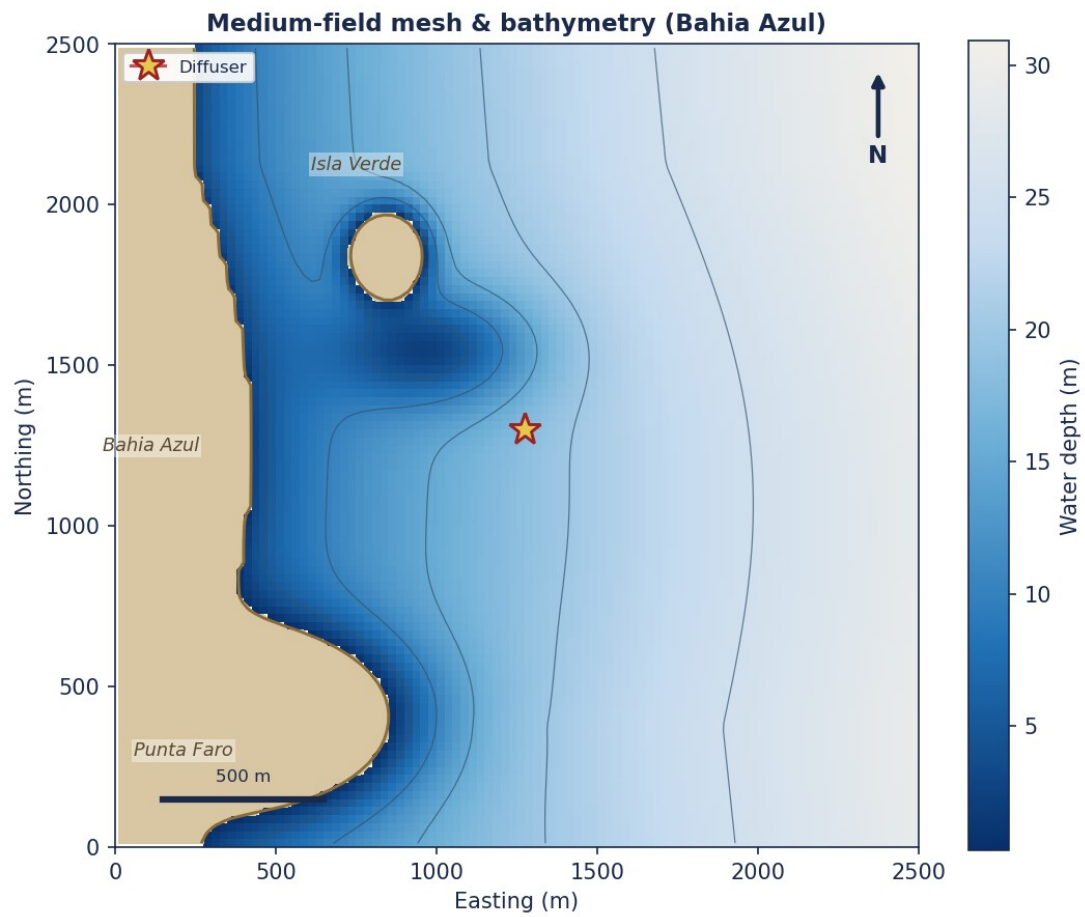


Figure. Medium-field mesh & bathymetry (Bahia Azul)

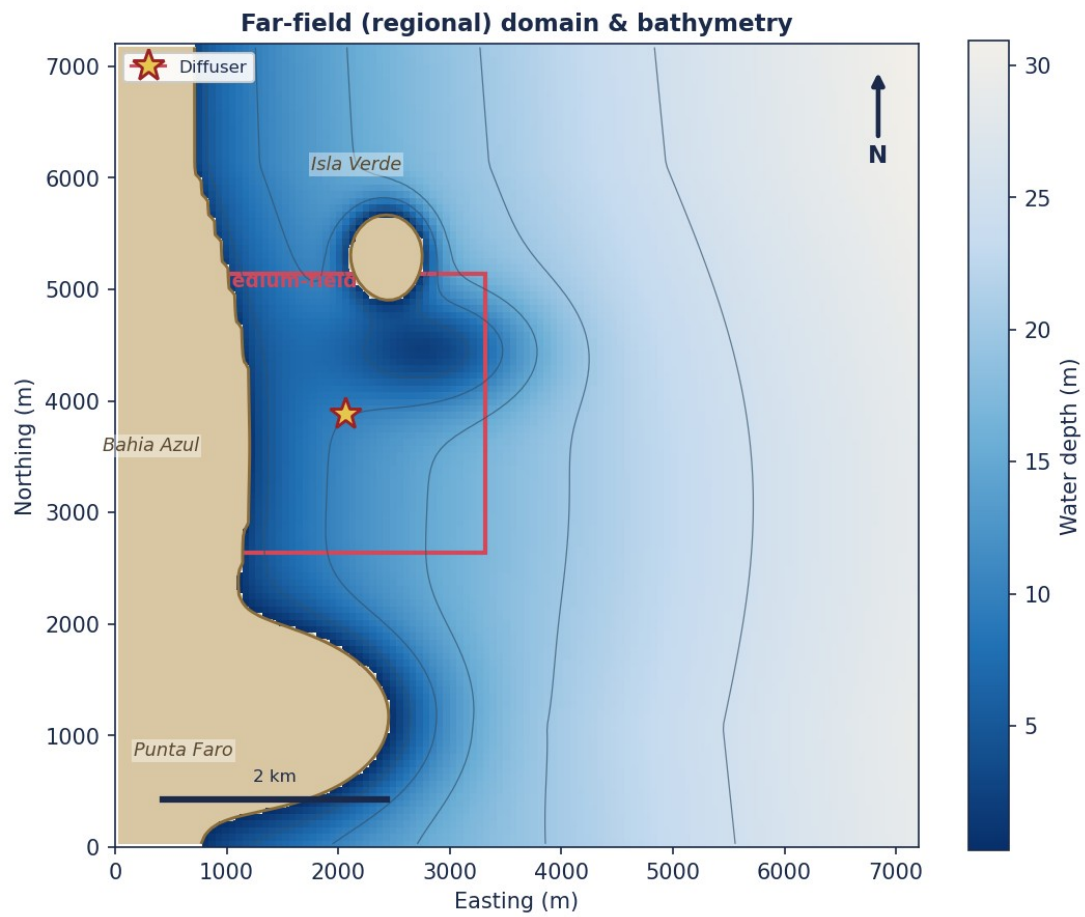


Figure. Far-field (regional) domain & bathymetry

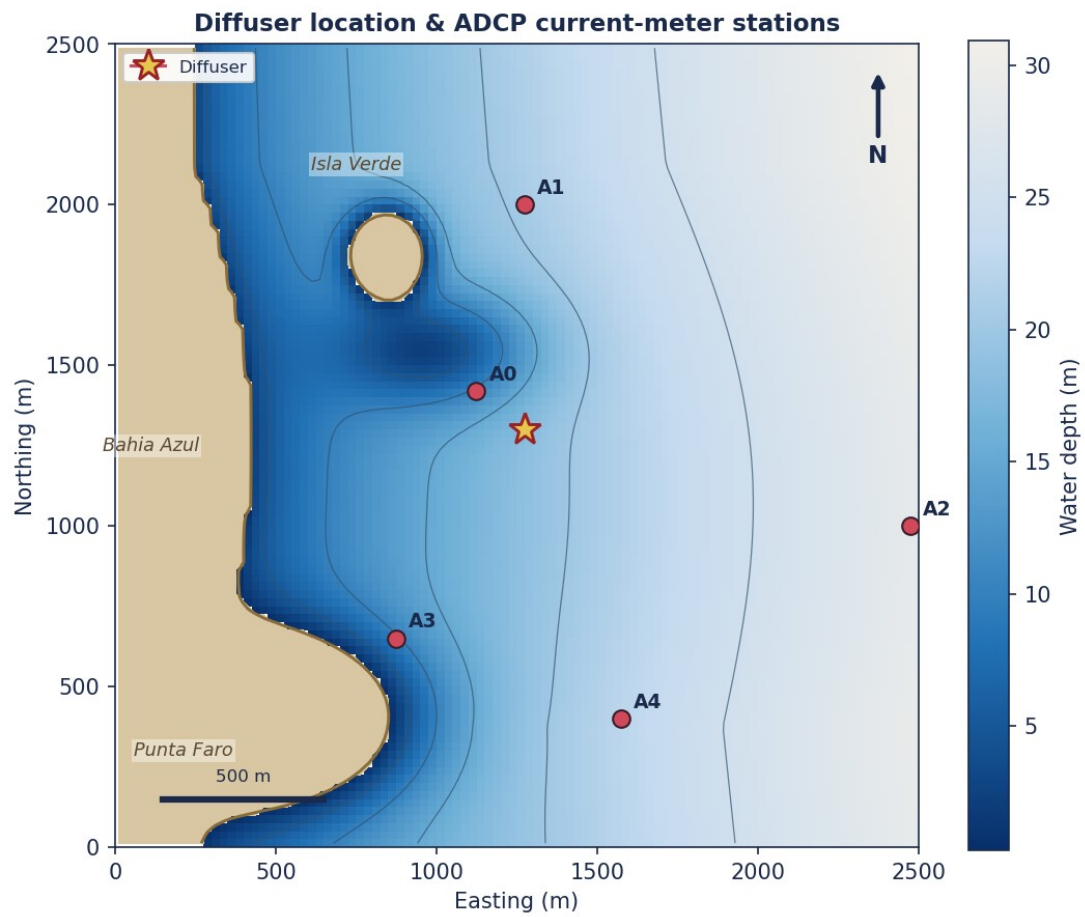


Figure. Diffuser location & ADCP current-meter stations

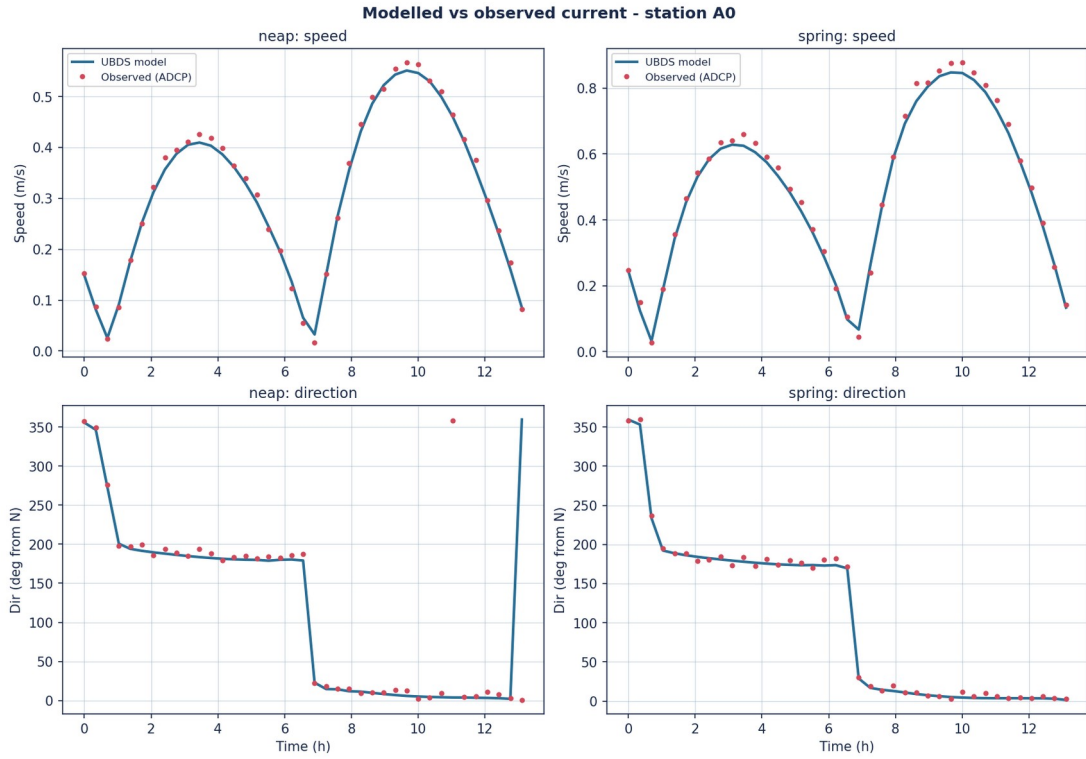


Figure. Modelled vs observed current speed & direction at station A0 (neap & spring).

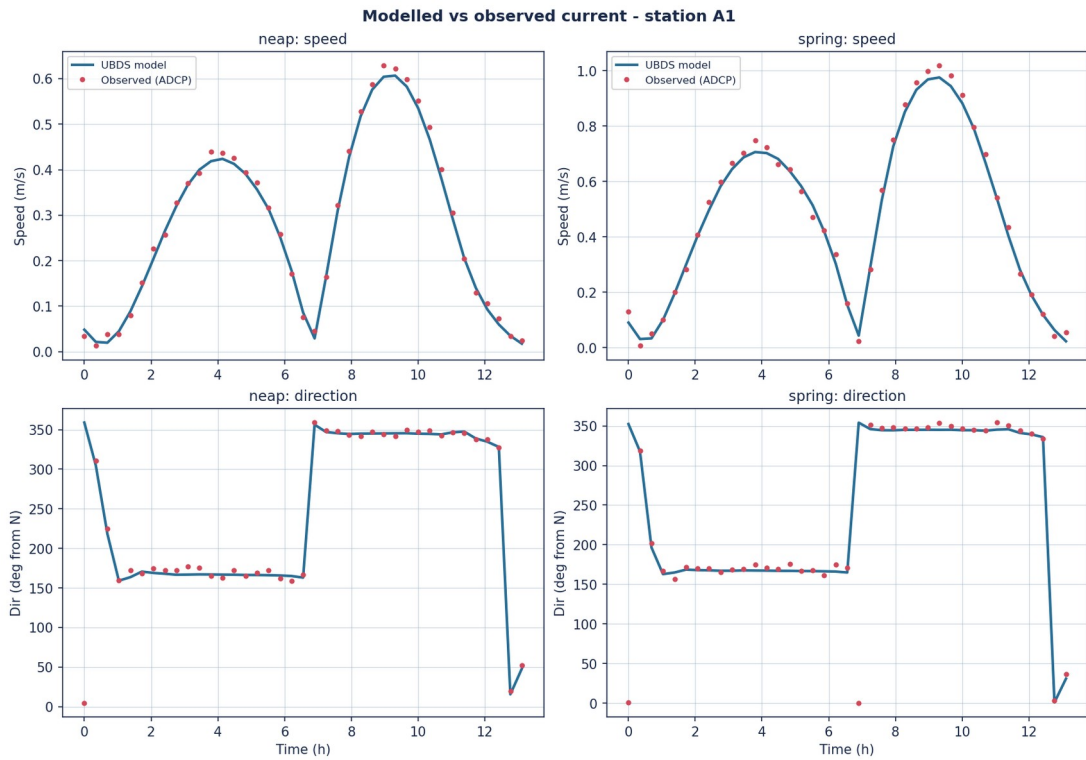


Figure. Modelled vs observed current speed & direction at station A1 (neap & spring).

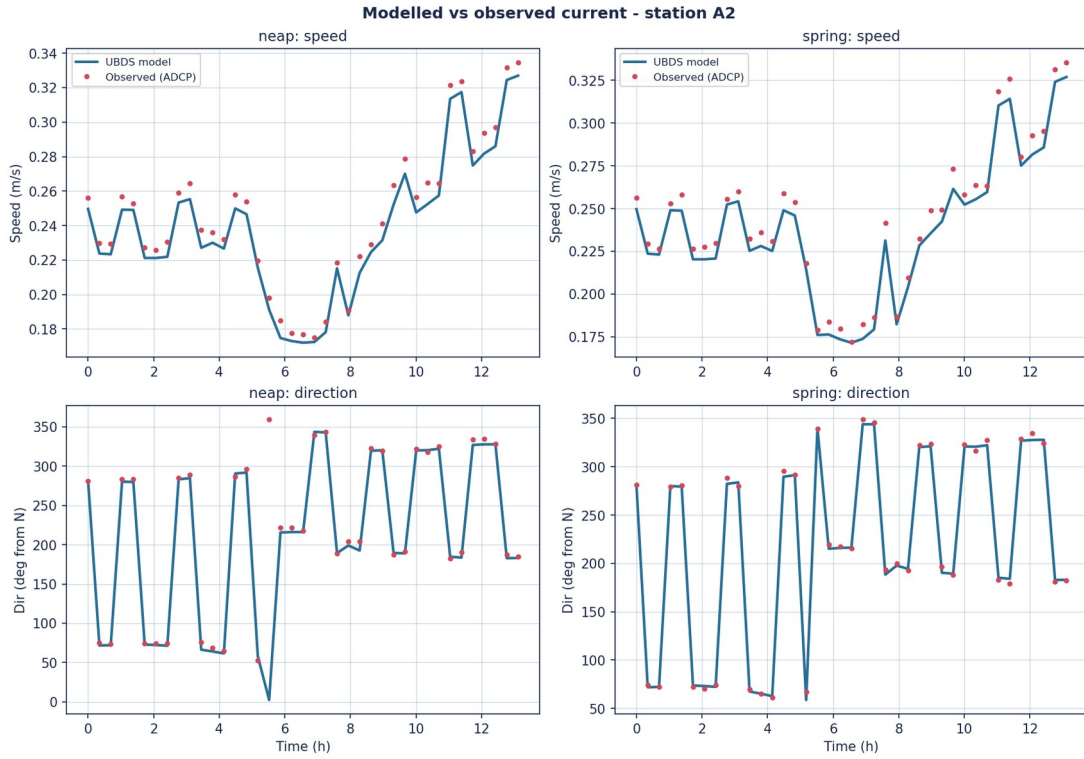


Figure. Modelled vs observed current speed & direction at station A2 (neap & spring).

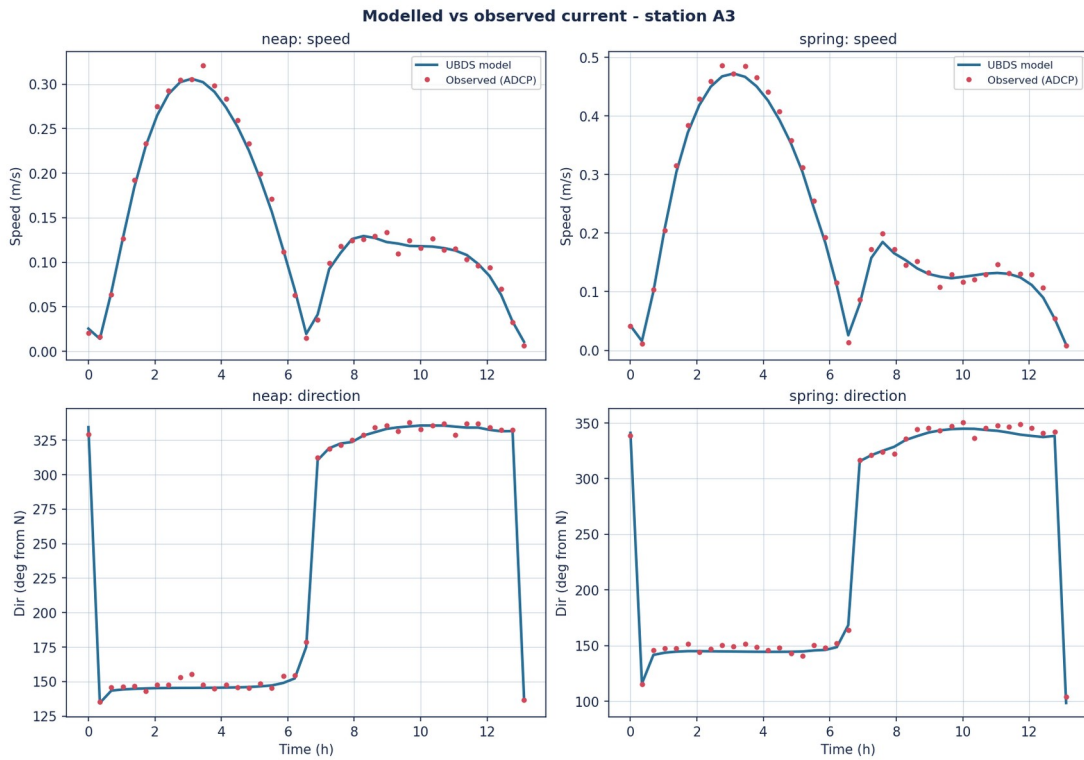


Figure. Modelled vs observed current speed & direction at station A3 (neap & spring).

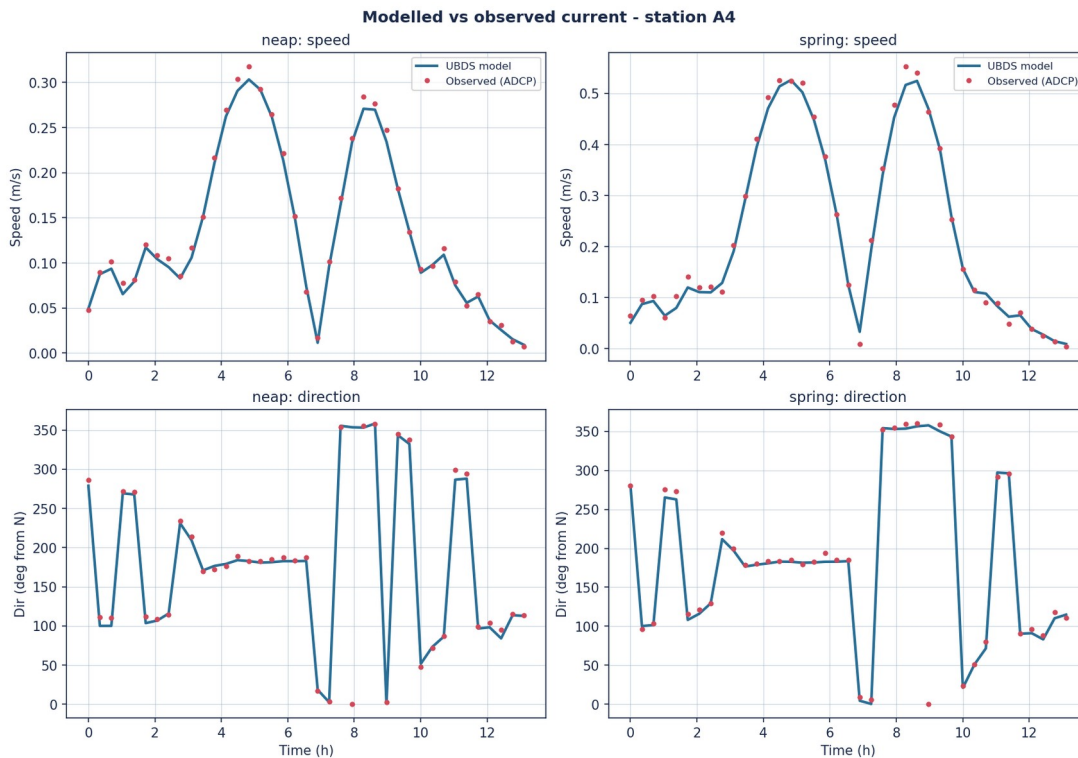


Figure. Modelled vs observed current speed & direction at station A4 (neap & spring).

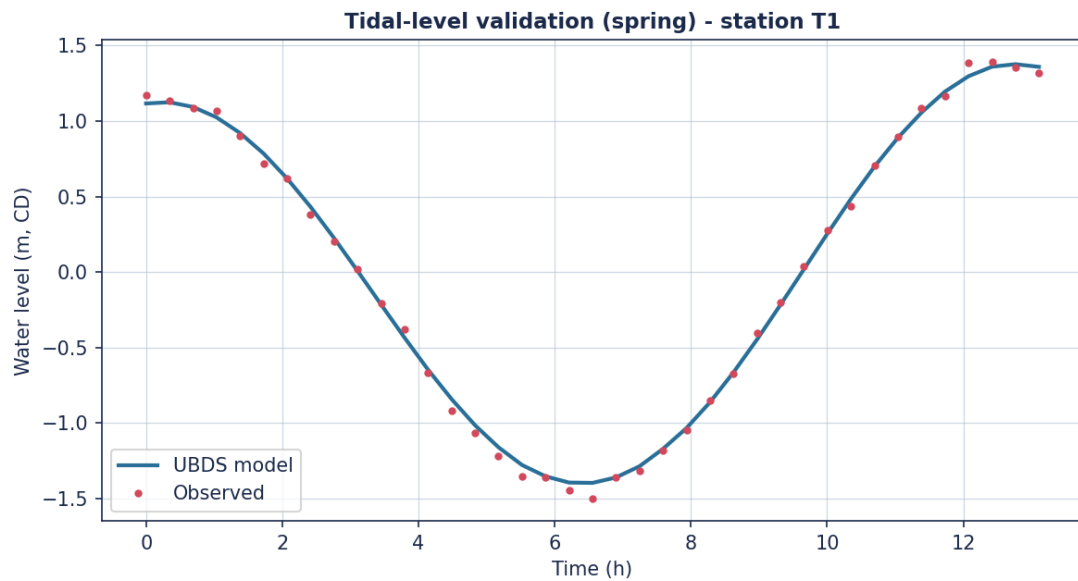


Figure. Modelled vs observed tidal level (spring tide) at the calibration station.

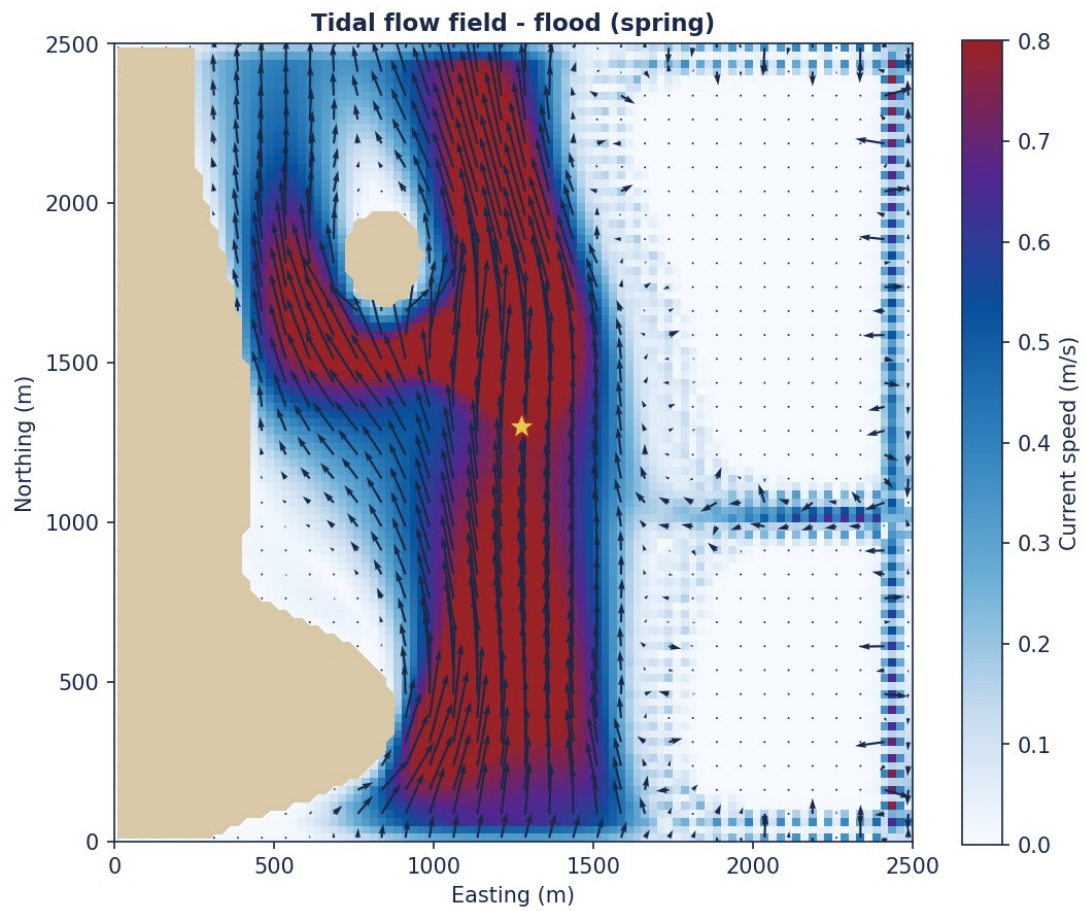


Figure. Depth-averaged tidal flow field at flood (spring tide).

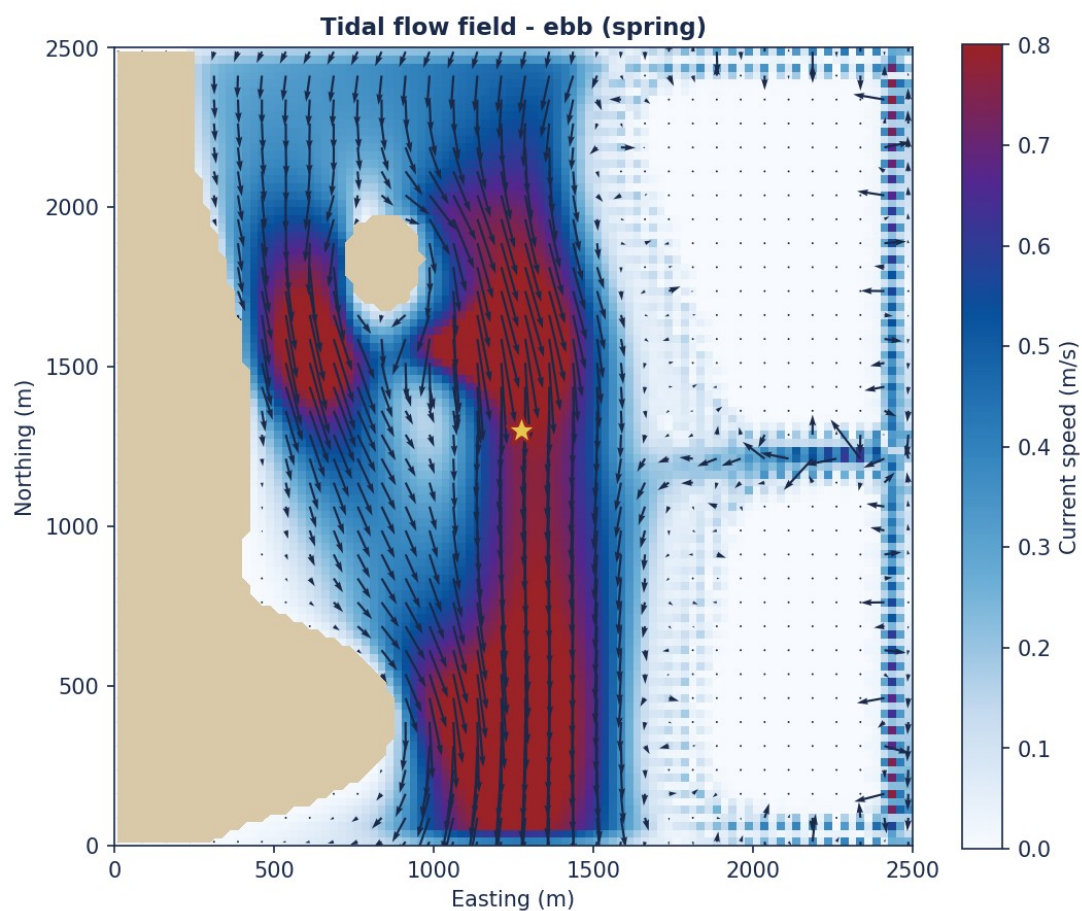


Figure. Depth-averaged tidal flow field at ebb (spring tide).

Table. Sigma-layer schemes (% of water-column depth per layer).

layer	pct_14	pct_9	pct_4
L14	40.0	40.0	35.0
L13	20.0	25.0	35.0
L12	11.0	15.0	15.0
L11	8.5	9.3	15.0
L10	6.5	4.5	nan
L9	4.5	2.5	nan
L8	3.0	1.5	nan
L7	2.0	1.2	nan
L6	1.5	1.0	nan
L5	1.0	nan	nan
L4	0.8	nan	nan
L3	0.6	nan	nan
L2	0.5	nan	nan
L1	0.1	nan	nan

Table. Hydrodynamic calibration skill at the ADCP stations.

station	tide	peak_speed_ms	rmse_speed_ms	willmott_d	nse
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A0	neap	0.551	0.011	0.999	0.996
A0	spring	0.848	0.02	0.998	0.993
A1	neap	0.607	0.013	0.999	0.995
A1	spring	0.975	0.024	0.998	0.994
A2	neap	0.327	0.008	0.992	0.967
A2	spring	0.327	0.007	0.992	0.968
A3	neap	0.306	0.007	0.998	0.994
A3	spring	0.472	0.011	0.999	0.995
A4	neap	0.303	0.007	0.999	0.995
A4	spring	0.526	0.014	0.999	0.994

4 Near-Field Initial Dilution

The near-field initial dilution was computed with the UBDS integral jet/plume engine for both brine types, all flow scenarios and a range of tidal velocities (slack, neap and spring).

Table. Near-field initial-dilution results.

brine	scenario	flow_m 3hr	n_ports	tide	velocity _ms	exit_vel ocity_ms	dilution	seabed _salinity_psu	rise_height_m	impact _distance_m
RO concent rate	S1	300	1	slack	0.08	4.57	16.9	38.36	5.48	4.0
RO concent rate	S1	300	1	neap	0.28	4.57	21.4	37.97	4.01	3.16
RO concent rate	S1	300	1	spring	0.55	4.57	24.2	37.8	3.17	3.73
RO concent rate	S2	700	1	slack	0.08	10.66	24.9	37.77	10.14	6.22
RO concent rate	S2	700	1	neap	0.28	10.66	28.0	37.63	7.01	4.42
RO concent rate	S2	700	1	spring	0.55	10.66	29.6	37.56	5.29	4.03
RO concent rate	S3	1200	2	slack	0.08	9.14	23.3	37.85	9.21	5.77
RO concent rate	S3	1200	2	neap	0.28	9.14	26.8	37.68	6.36	4.12
RO concent rate	S3	1200	2	spring	0.55	9.14	28.9	37.59	4.85	3.96
Saturat ed cavern brine	S1	300	1	slack	0.08	4.57	14.3	51.4	2.65	3.35
Saturat ed cavern brine	S1	300	1	neap	0.28	4.57	28.7	43.95	2.36	3.58

Saturat ed cavern brine	S1	300	1	spring	0.55	4.57	44.3	41.32	2.11	5.95
Saturat ed cavern brine	S2	700	1	slack	0.08	10.66	31.6	43.27	5.7	8.14
Saturat ed cavern brine	S2	700	1	neap	0.28	10.66	16.7	49.3	4.99	3.73
Saturat ed cavern brine	S2	700	1	spring	0.55	10.66	18.7	47.89	4.24	3.19
Saturat ed cavern brine	S3	1200	2	slack	0.08	9.14	27.1	44.39	4.93	6.92
Saturat ed cavern brine	S3	1200	2	neap	0.28	9.14	52.8	40.54	4.34	7.33
Saturat ed cavern brine	S3	1200	2	spring	0.55	9.14	17.5	48.72	3.75	2.97

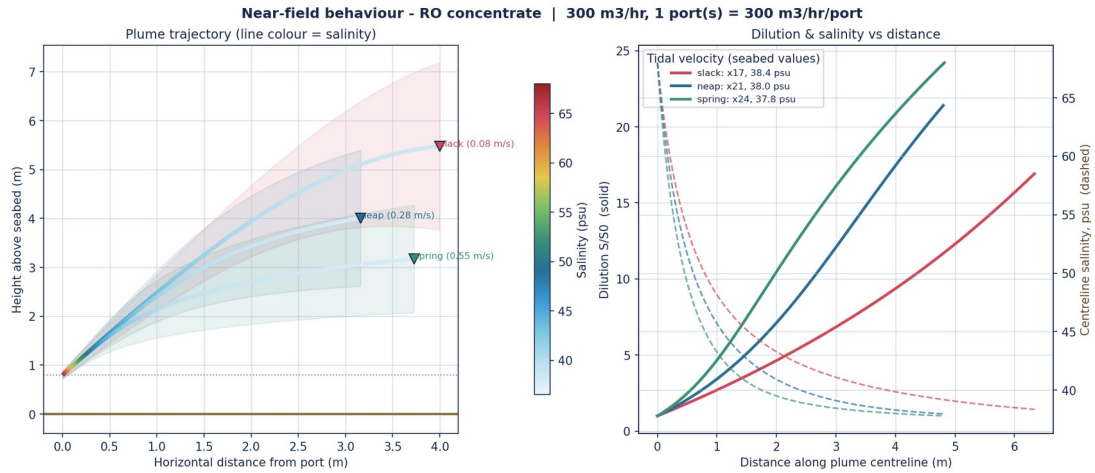


Figure. Near-field trajectory, dilution and salinity - RO concentrate, 300 m3/hr (300 m3/hr per port).

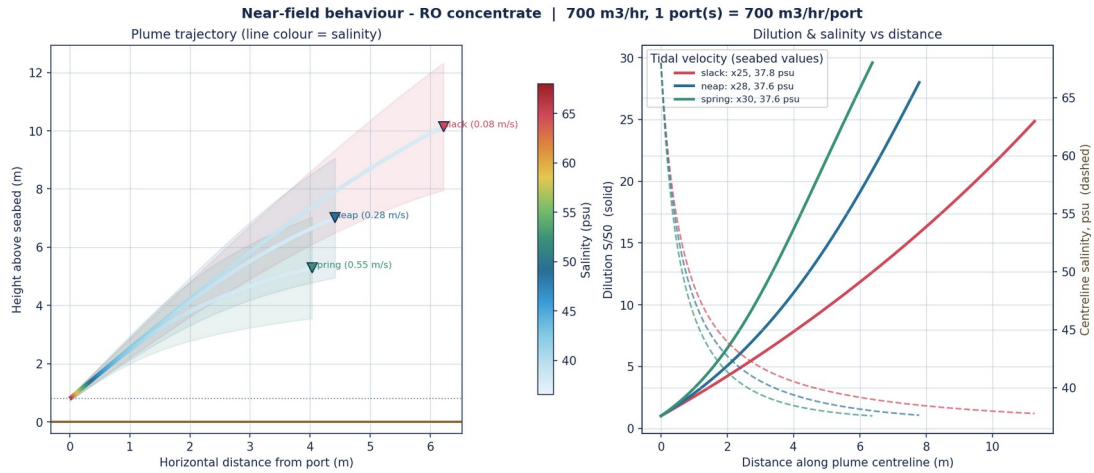


Figure. Near-field trajectory, dilution and salinity - RO concentrate, 700 m³/hr (700 m³/hr per port).

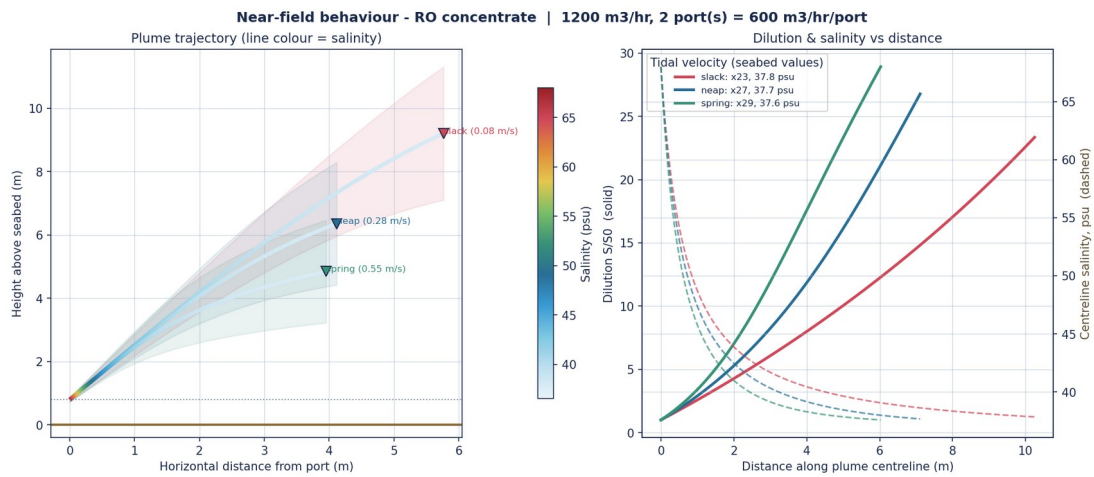


Figure. Near-field trajectory, dilution and salinity - RO concentrate, 1200 m³/hr (600 m³/hr per port).

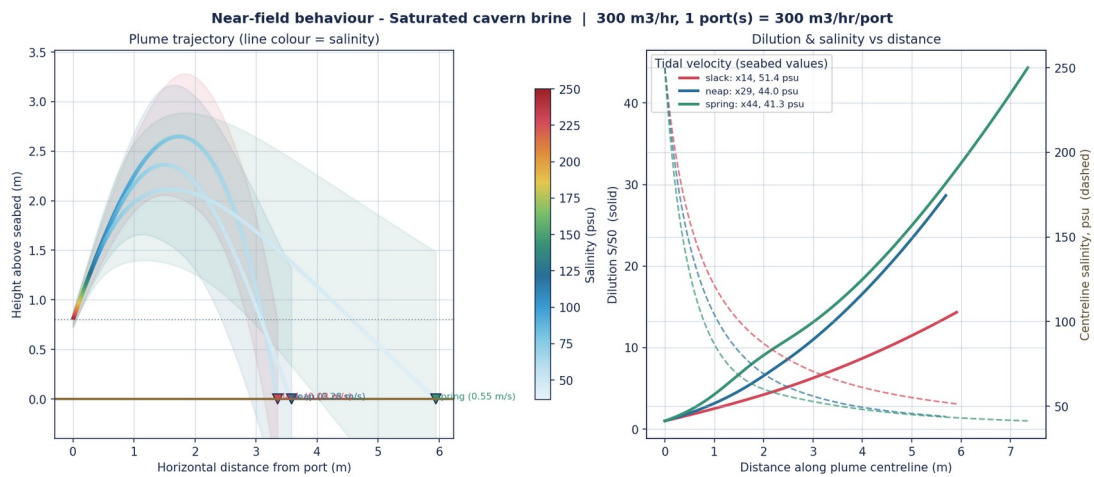


Figure. Near-field trajectory, dilution and salinity - Saturated cavern brine, 300 m³/hr (300 m³/hr per port).

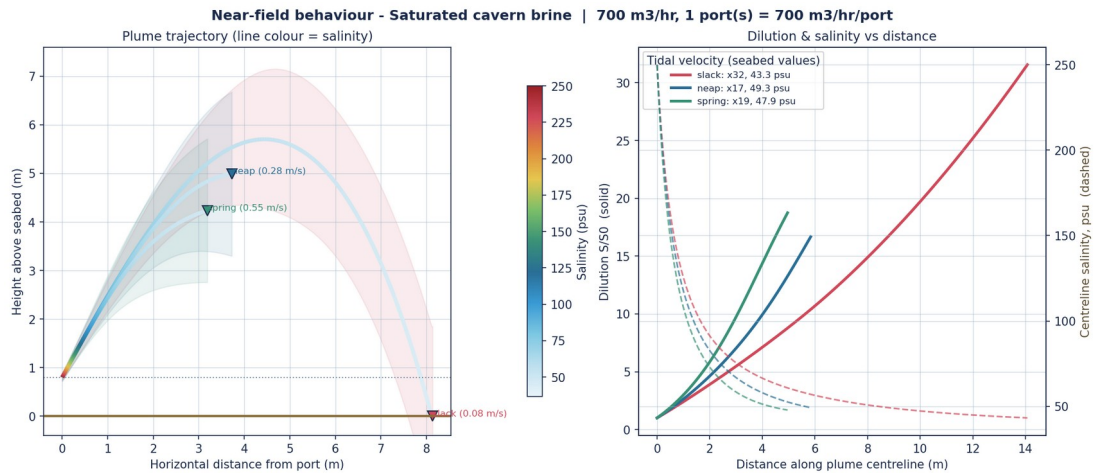


Figure. Near-field trajectory, dilution and salinity - Saturated cavern brine, 700 m³/hr (700 m³/hr per port).

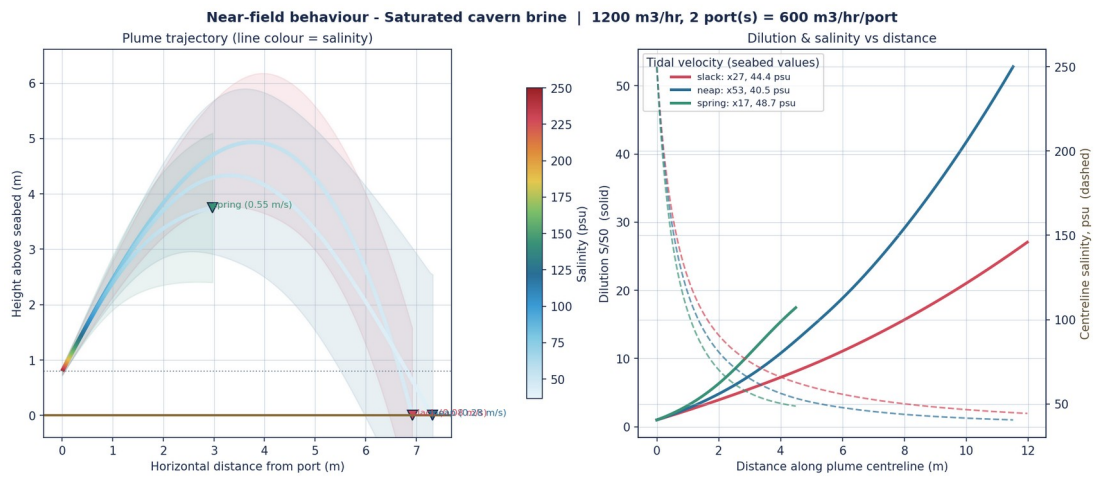


Figure. Near-field trajectory, dilution and salinity - Saturated cavern brine, 1200 m³/hr (600 m³/hr per port).

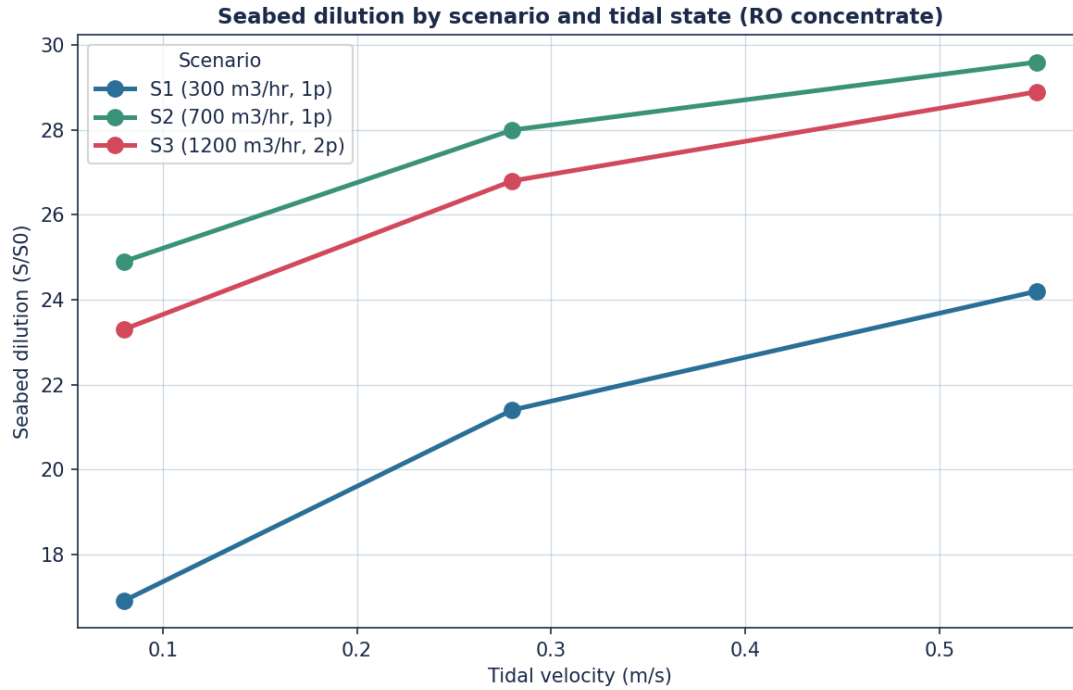


Figure. Seabed initial dilution by flow scenario and tidal state.

5 Medium-Field Dispersion

The medium-field dispersion was simulated with the sigma-layer transport engine driven by the calibrated tidal currents, including a layer-resolution sensitivity test and the near-field-coupled brine source. Maximum-salinity envelopes, tidal-phase snapshots with current vectors, and vertical profiles through the diffuser are presented.

Layer-resolution sensitivity (maximum seabed salinity, psu): 4-layer = 41.5, 9-layer = 43.4, 14-layer = 44.8. The 9-layer scheme was adopted for the production runs.

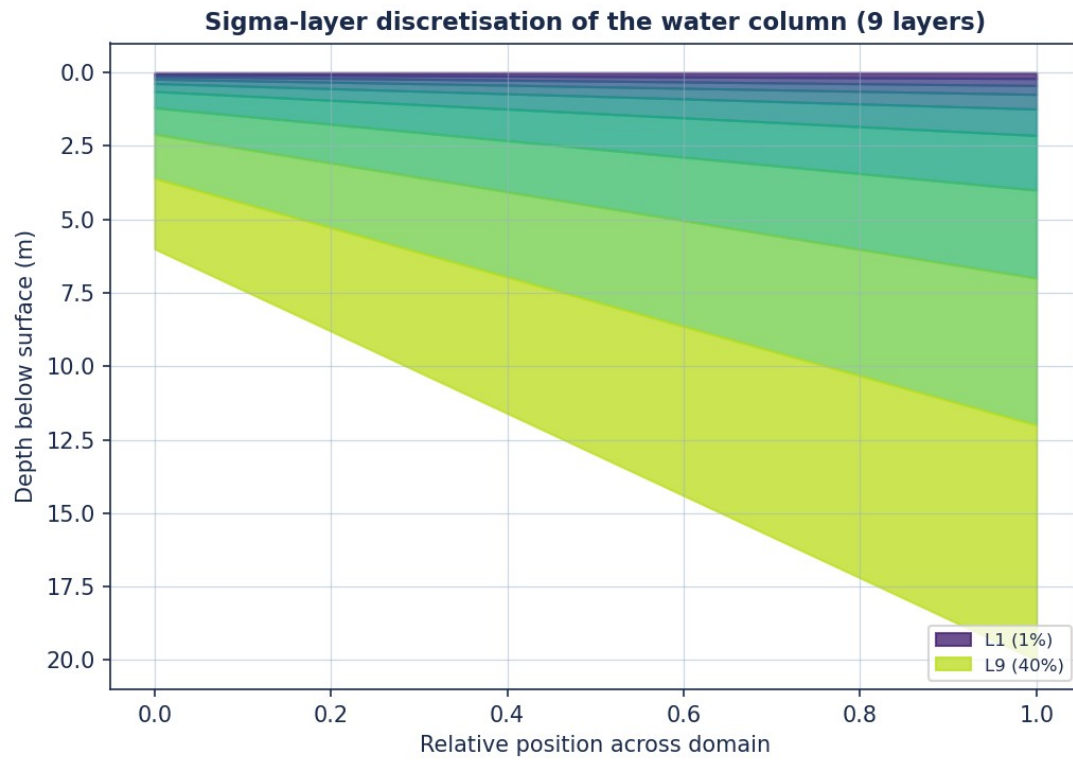


Figure. Variation of the terrain-following sigma layers with water depth (9-layer scheme).

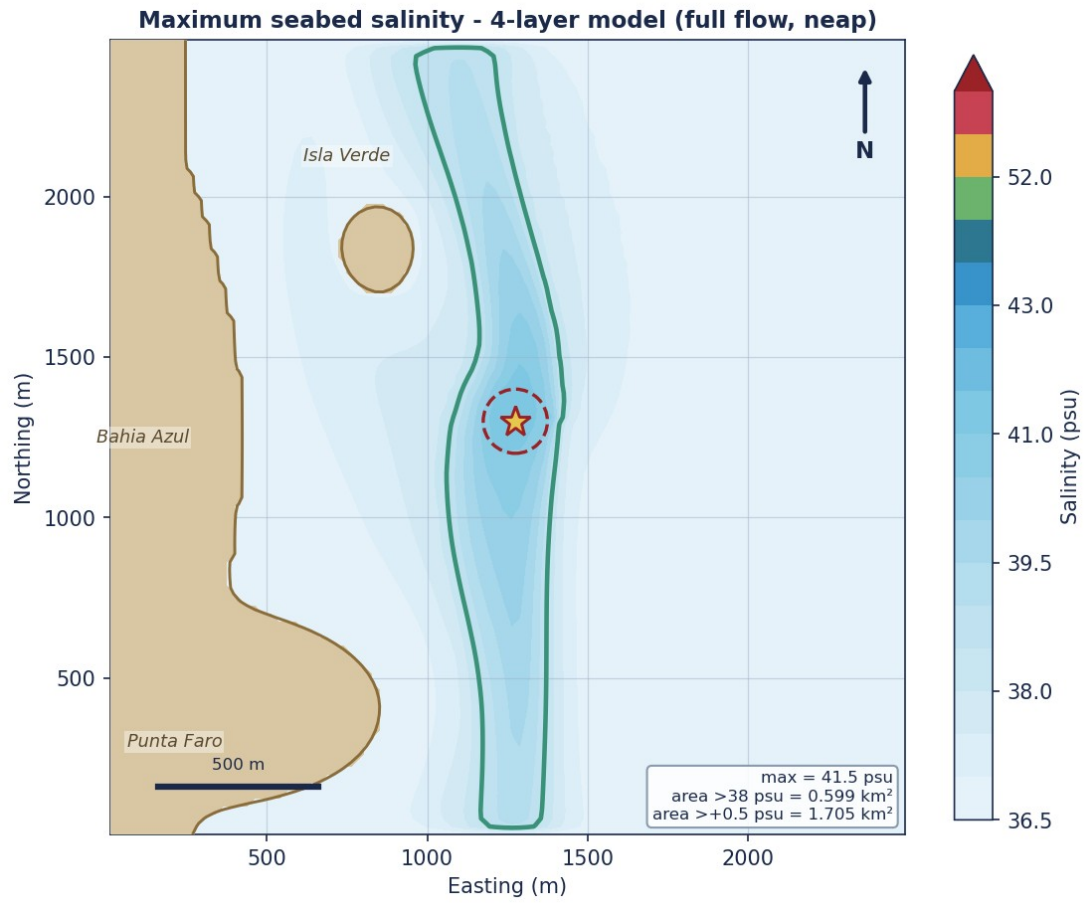


Figure. Maximum seabed salinity - 4-layer model (full flow, neap)

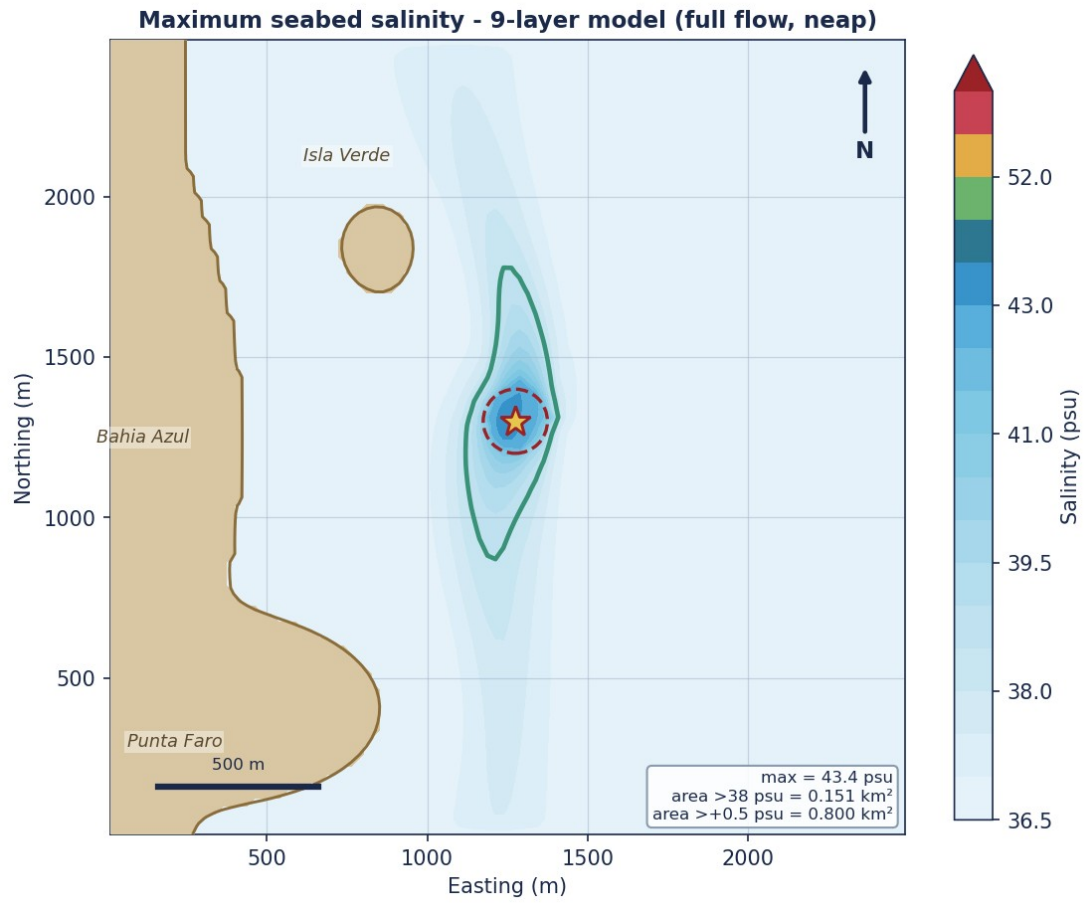


Figure. Maximum seabed salinity - 9-layer model (full flow, neap)

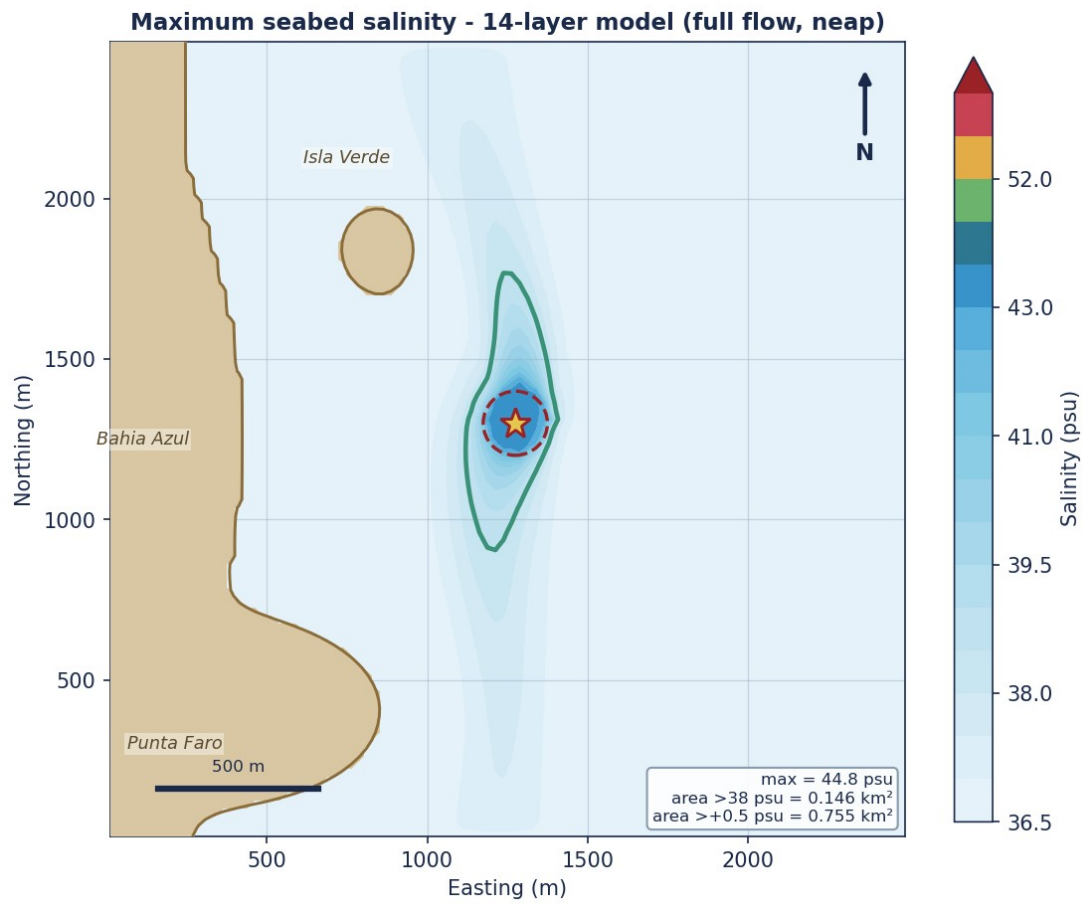


Figure. Maximum seabed salinity - 14-layer model (full flow, neap)

Maximum salinity vs height above the seabed - neap, full flow (dense plume is bottom-trapped and decays upward)

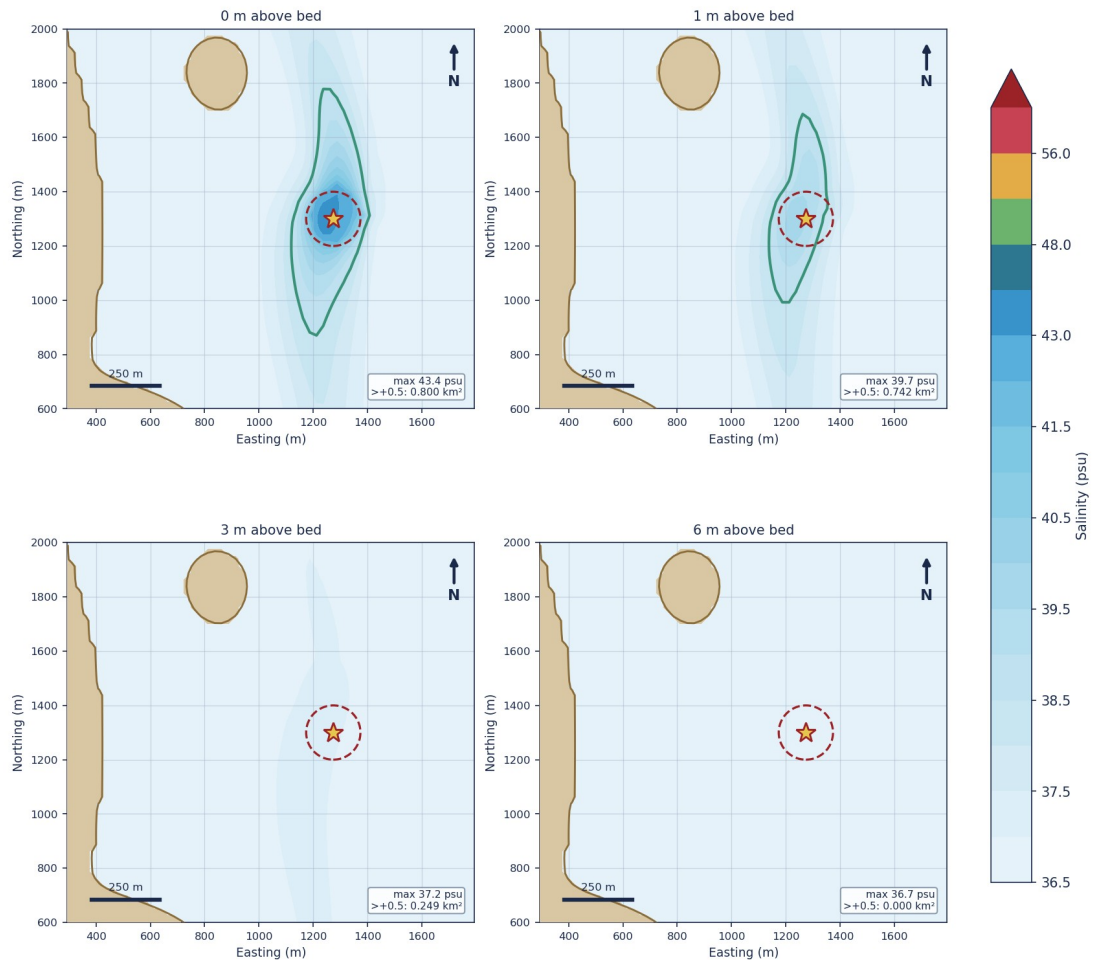


Figure. Maximum salinity vs height above the seabed - neap, full flow (dense plume is bottom-trapped and decays upward)

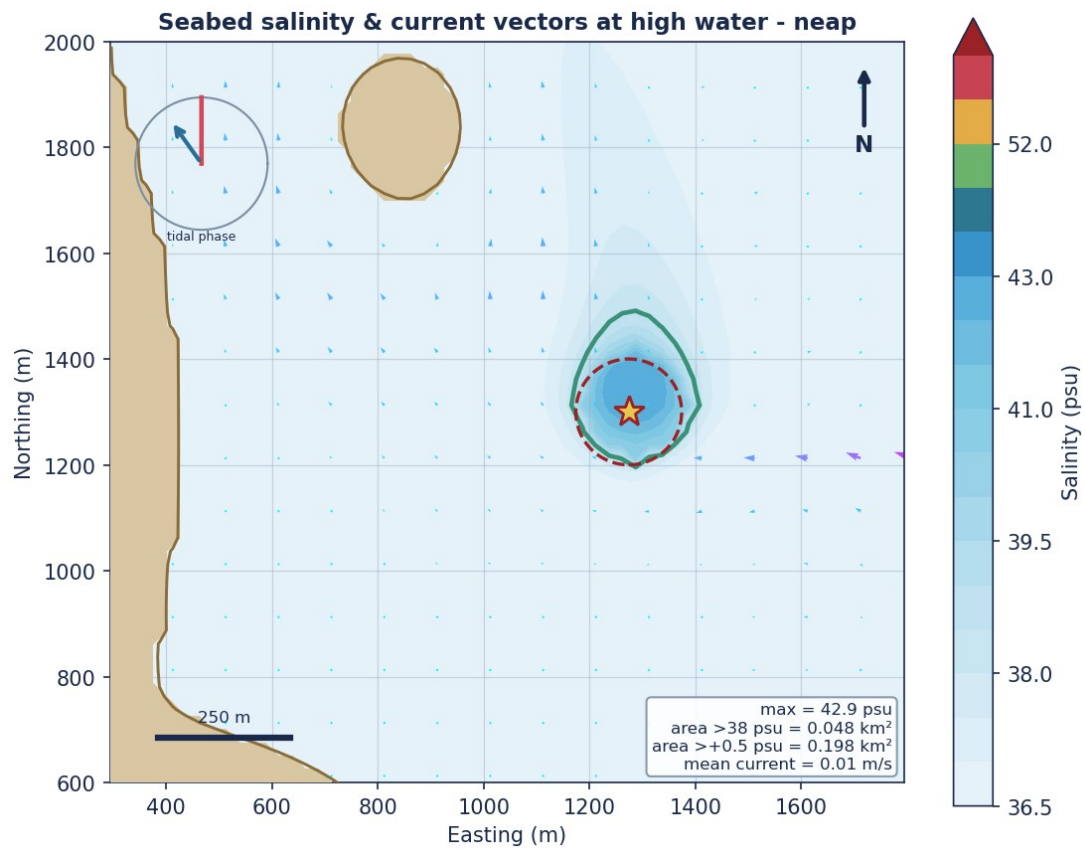


Figure. Seabed salinity & current vectors at high water - neap

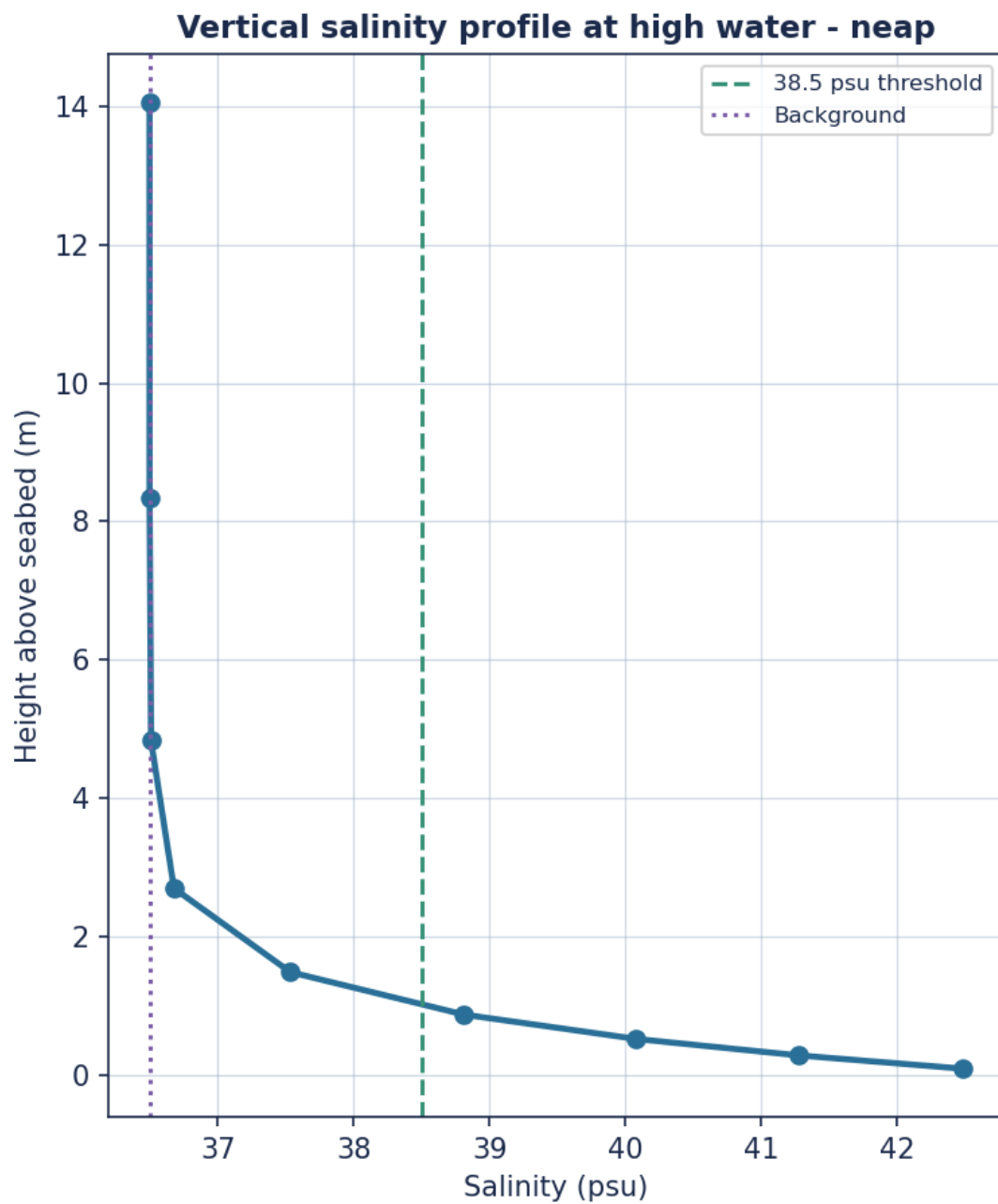


Figure. Vertical salinity profile at high water - neap

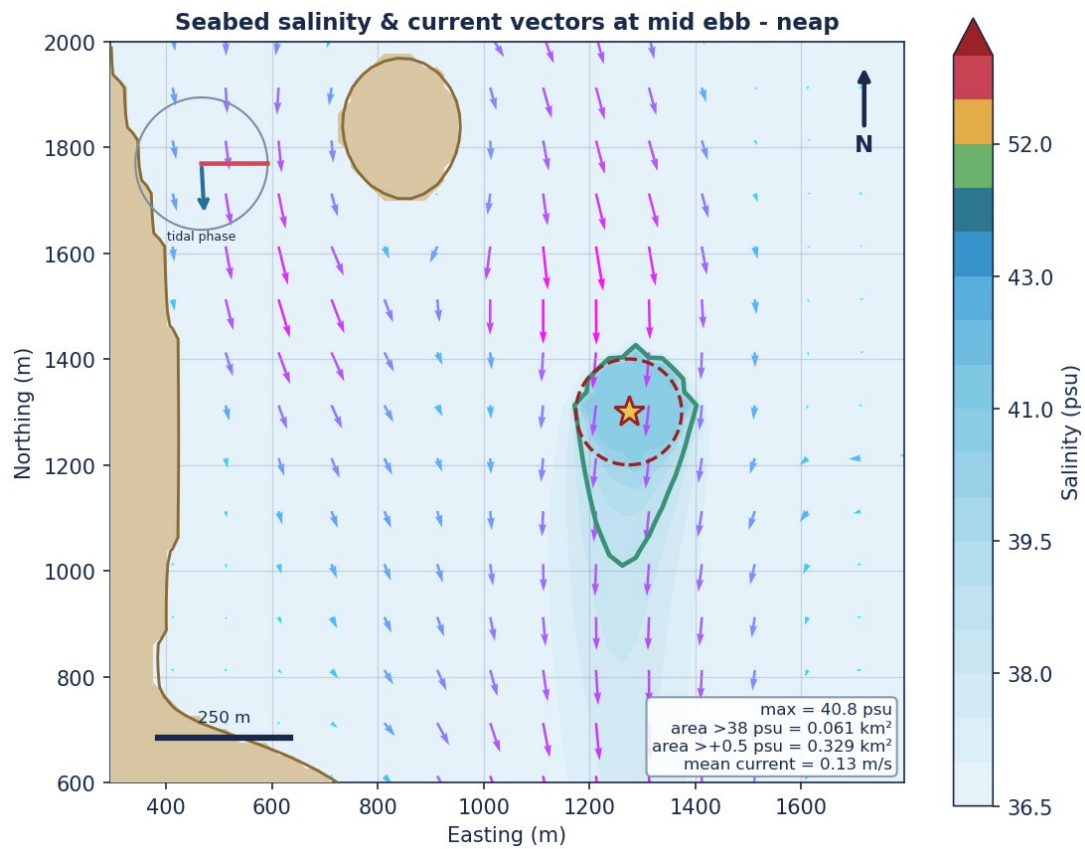


Figure. Seabed salinity & current vectors at mid ebb - neap

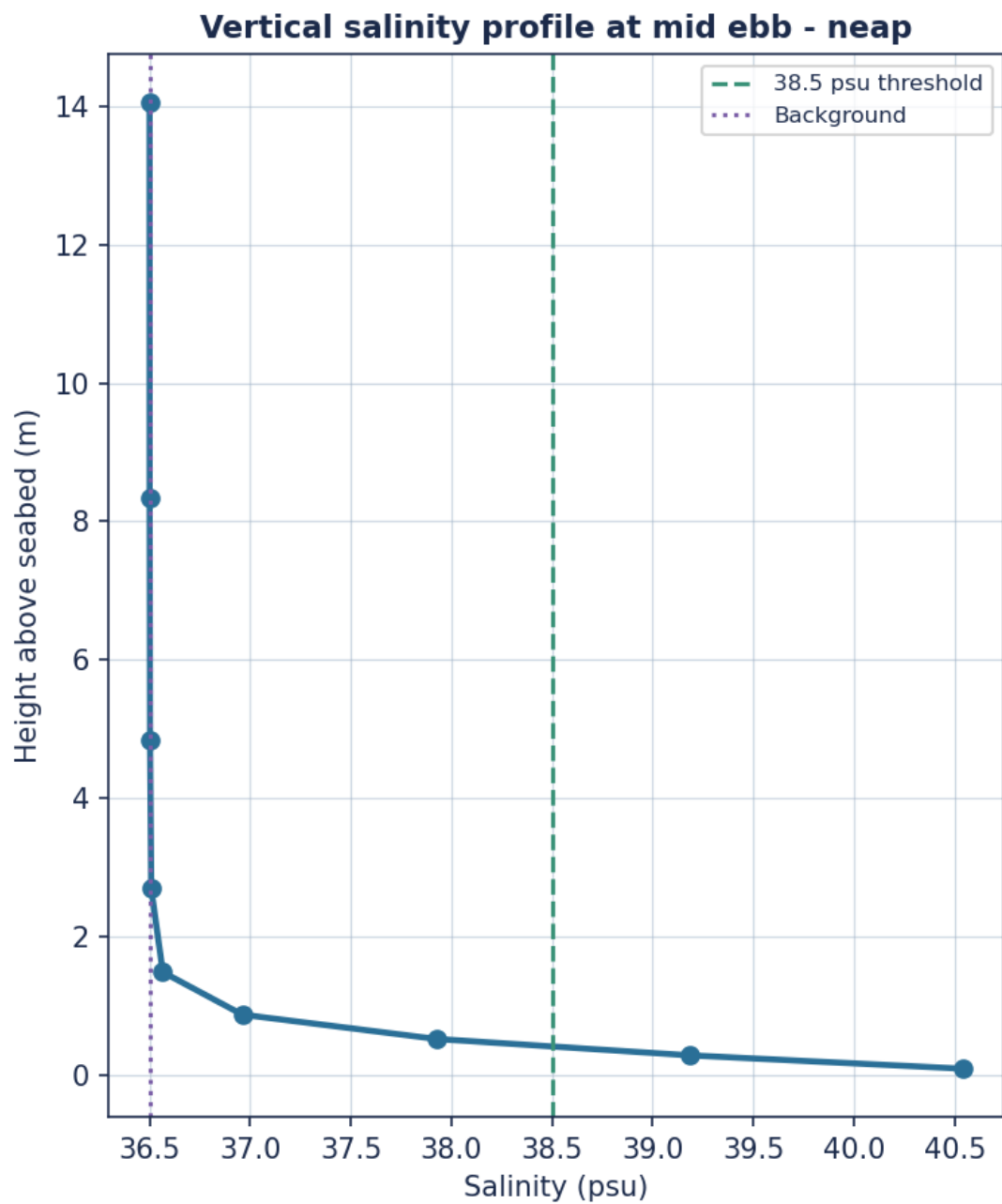


Figure. Vertical salinity profile at mid ebb - neap

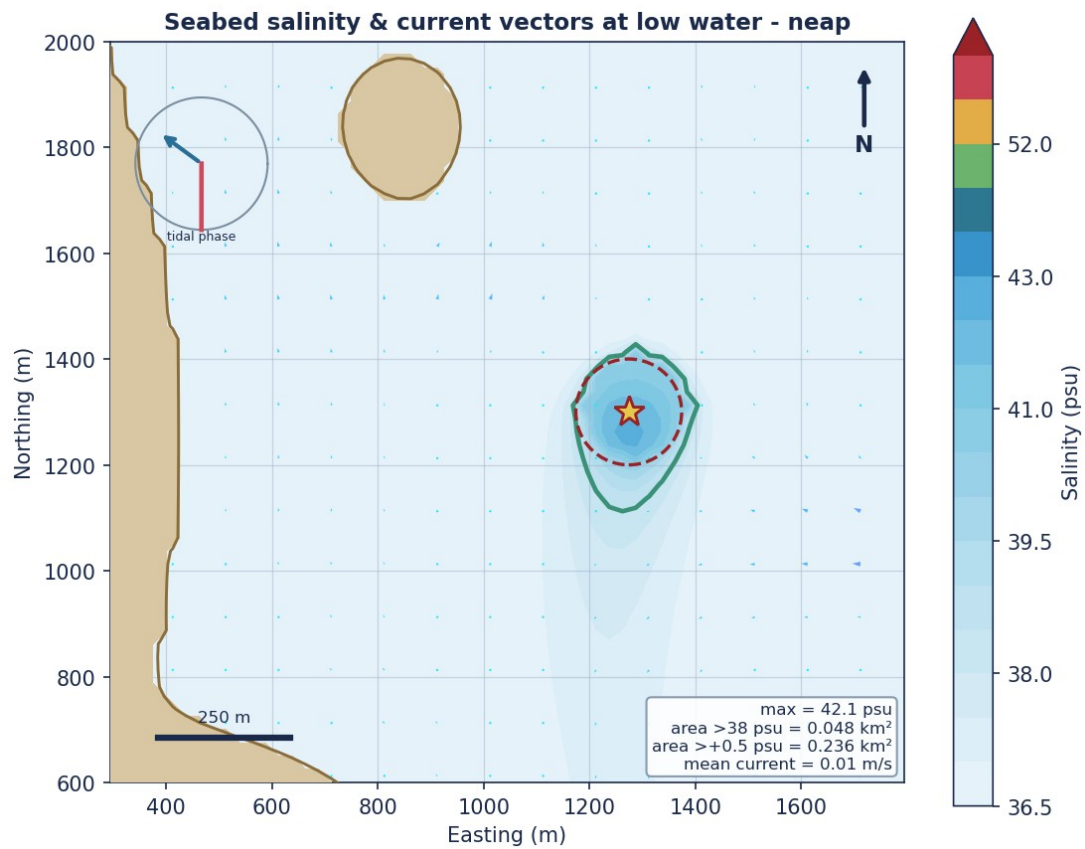


Figure. Seabed salinity & current vectors at low water - neap

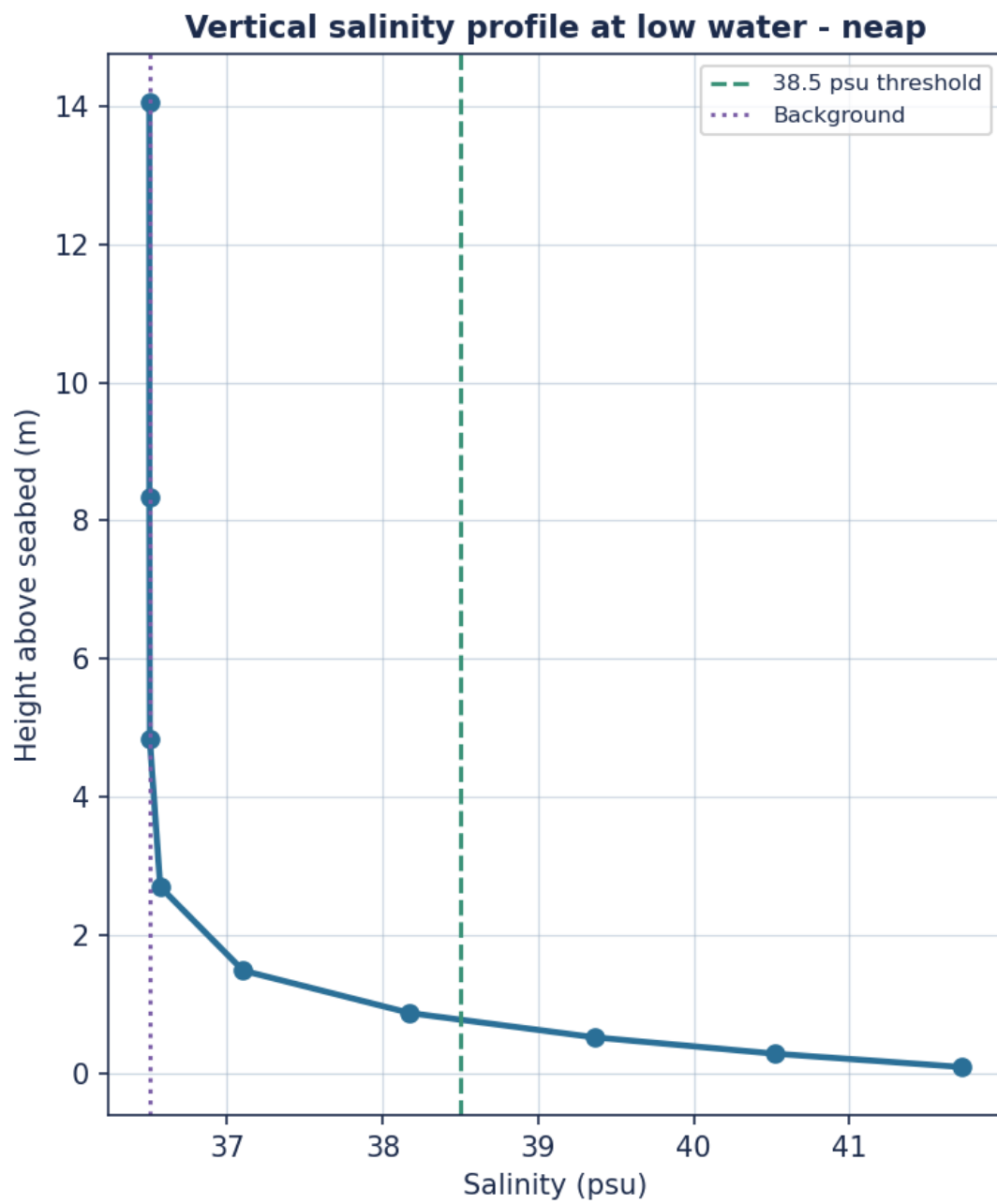


Figure. Vertical salinity profile at low water - neap

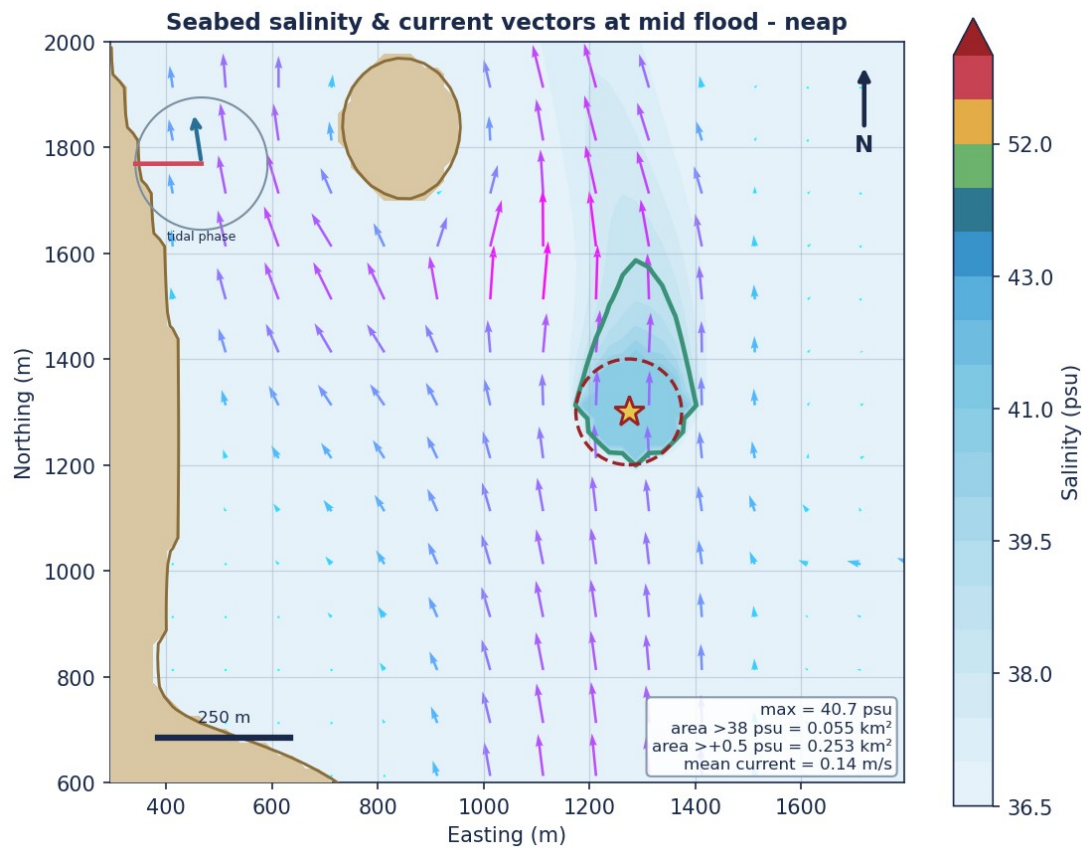


Figure. Seabed salinity & current vectors at mid flood - neap

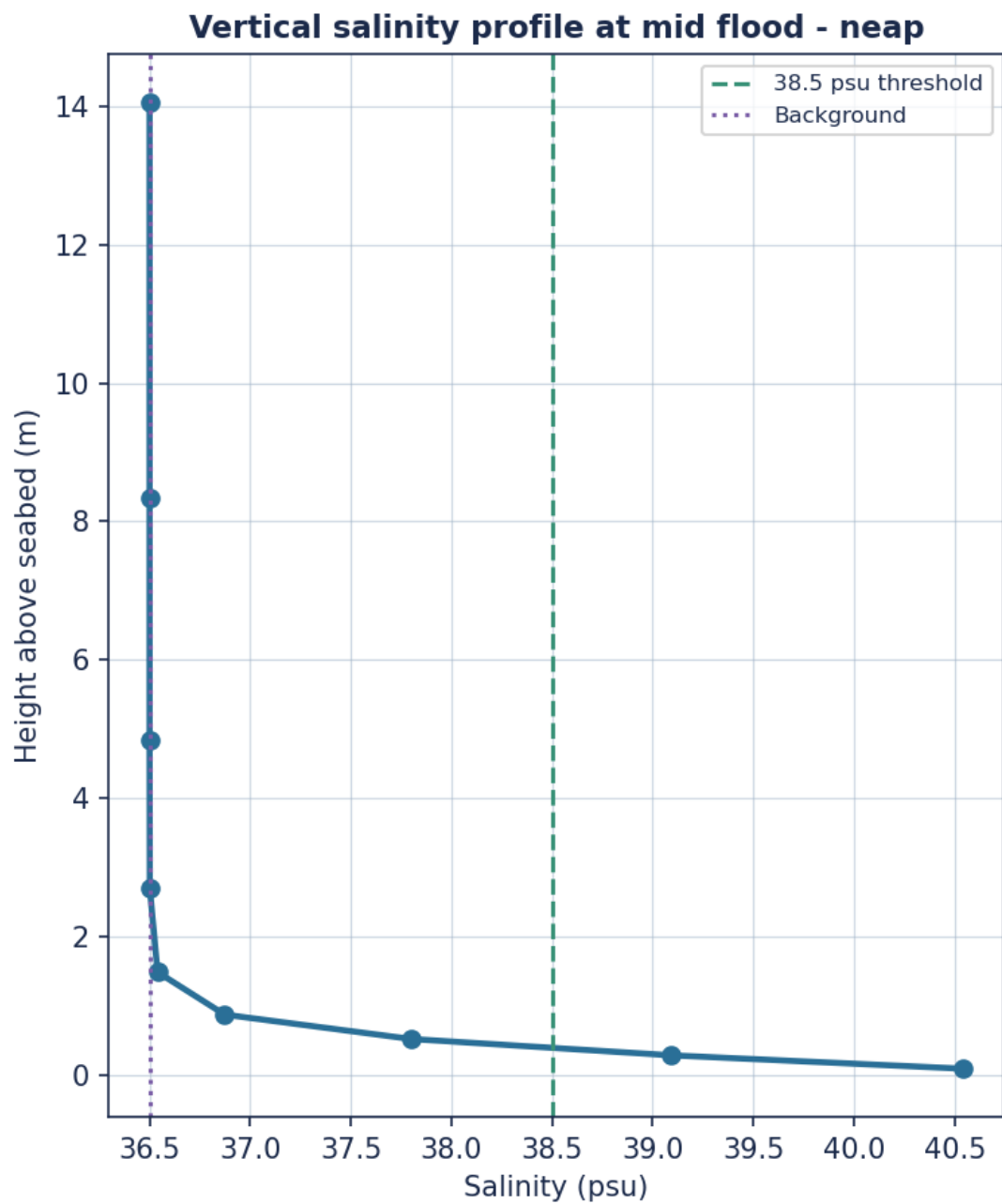


Figure. Vertical salinity profile at mid flood - neap

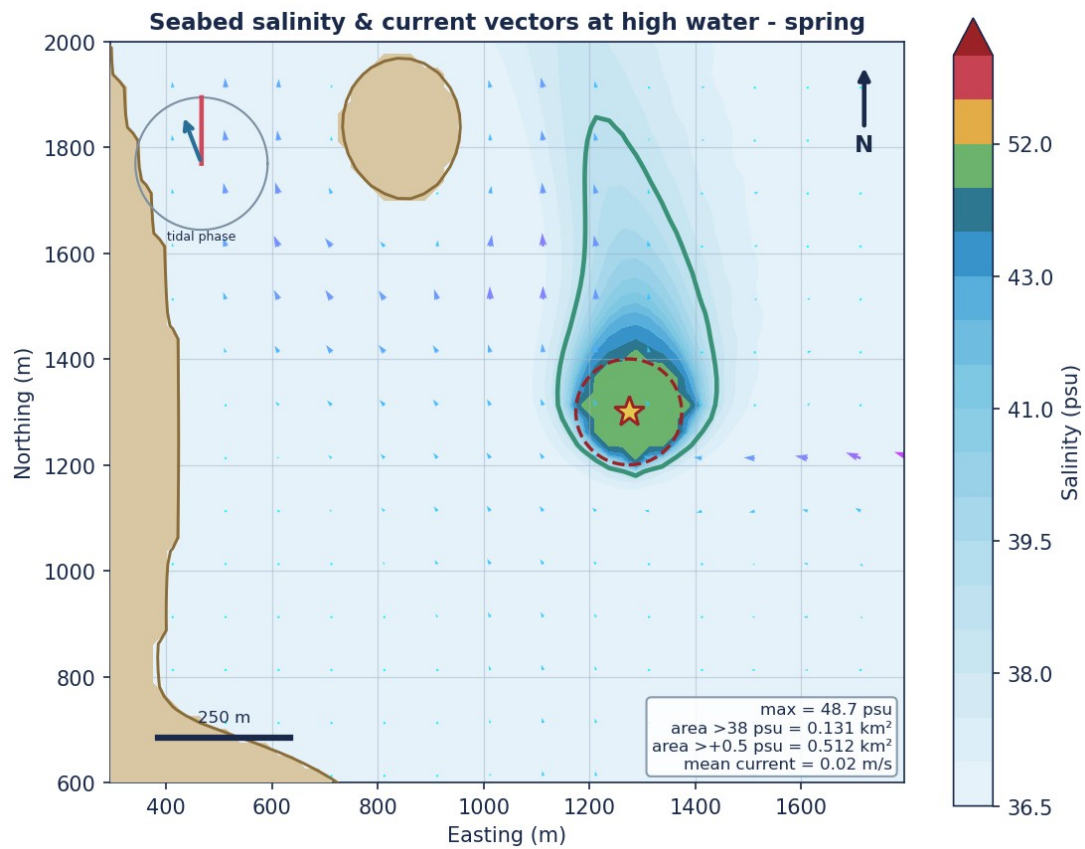


Figure. Seabed salinity & current vectors at high water - spring

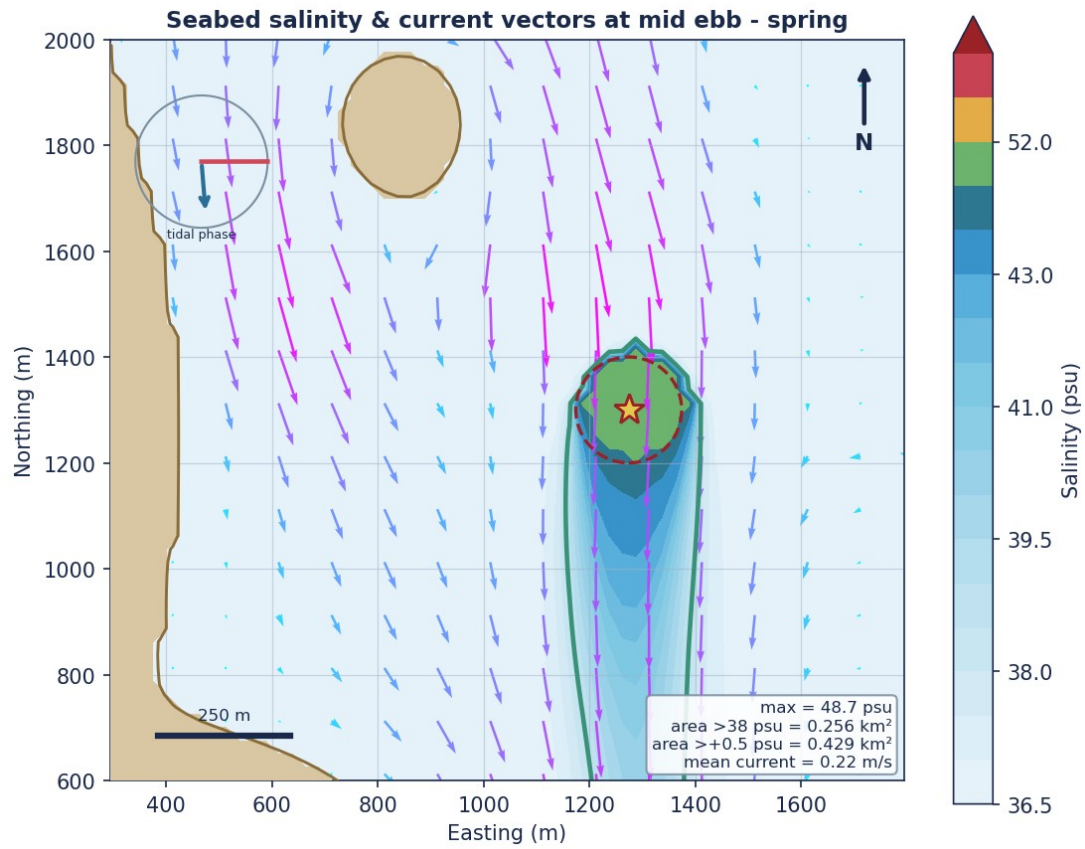


Figure. Seabed salinity & current vectors at mid ebb - spring

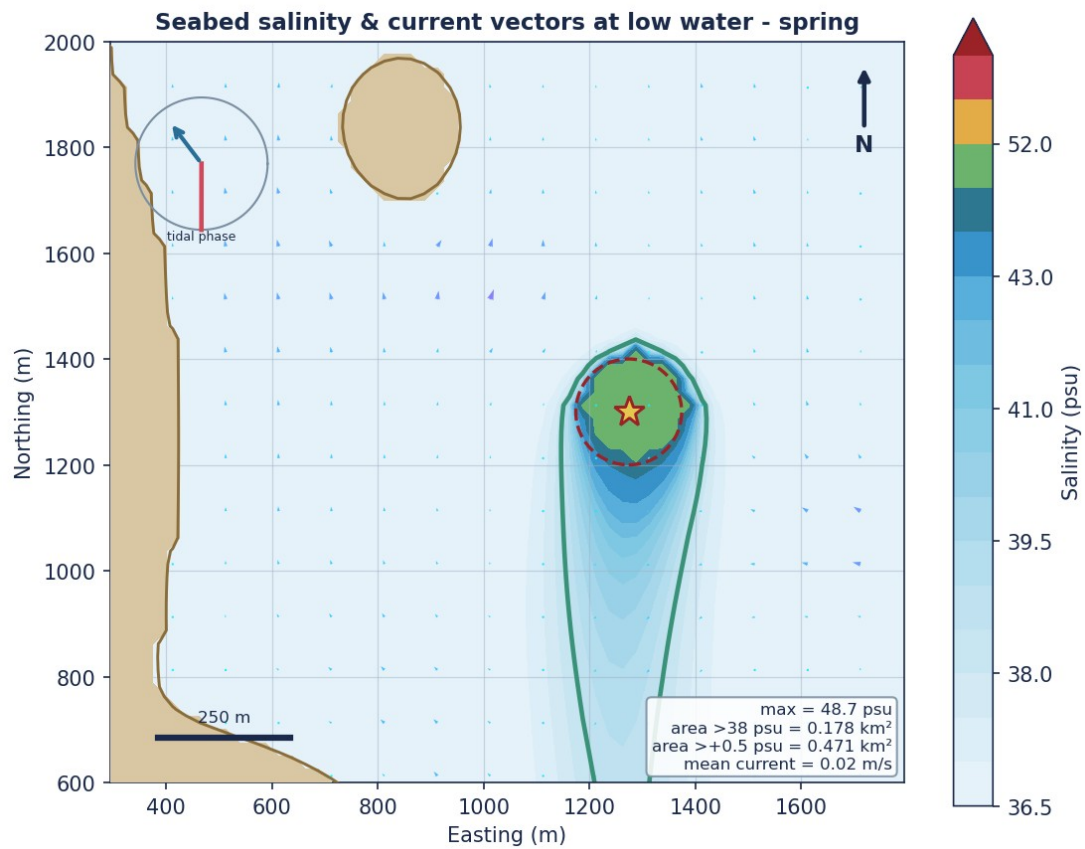


Figure. Seabed salinity & current vectors at low water - spring

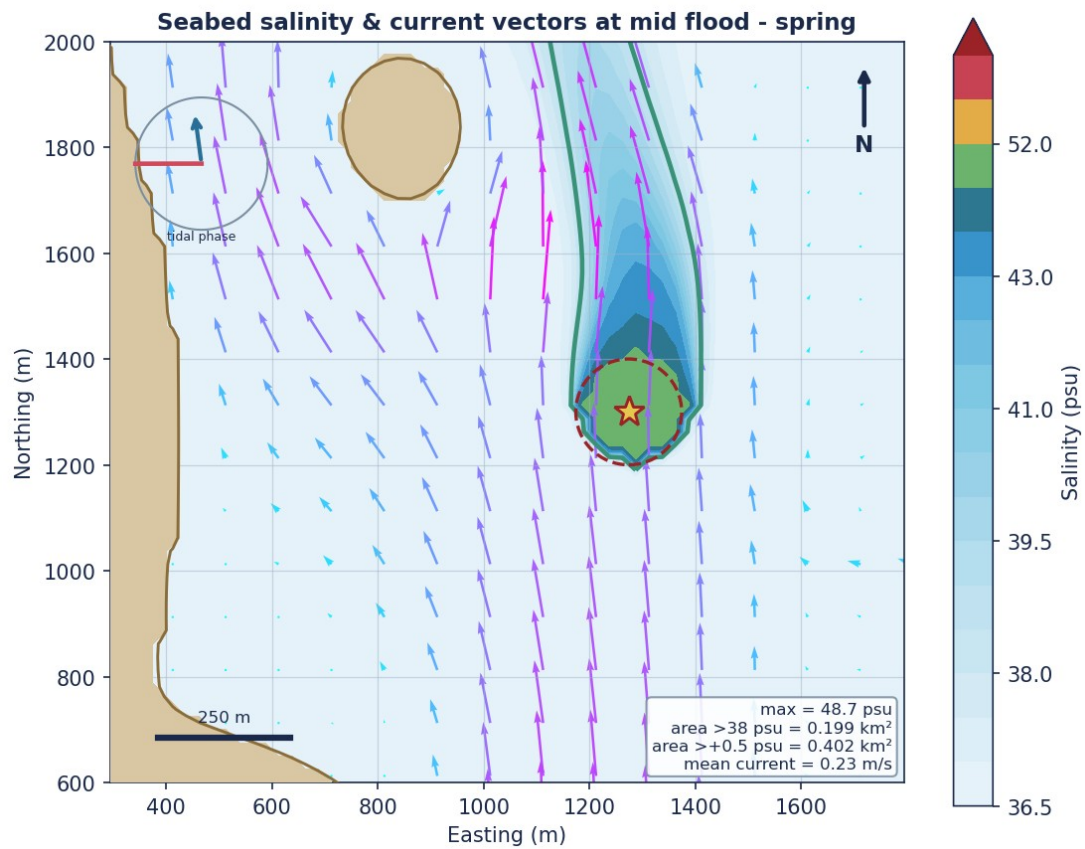


Figure. Seabed salinity & current vectors at mid flood - spring

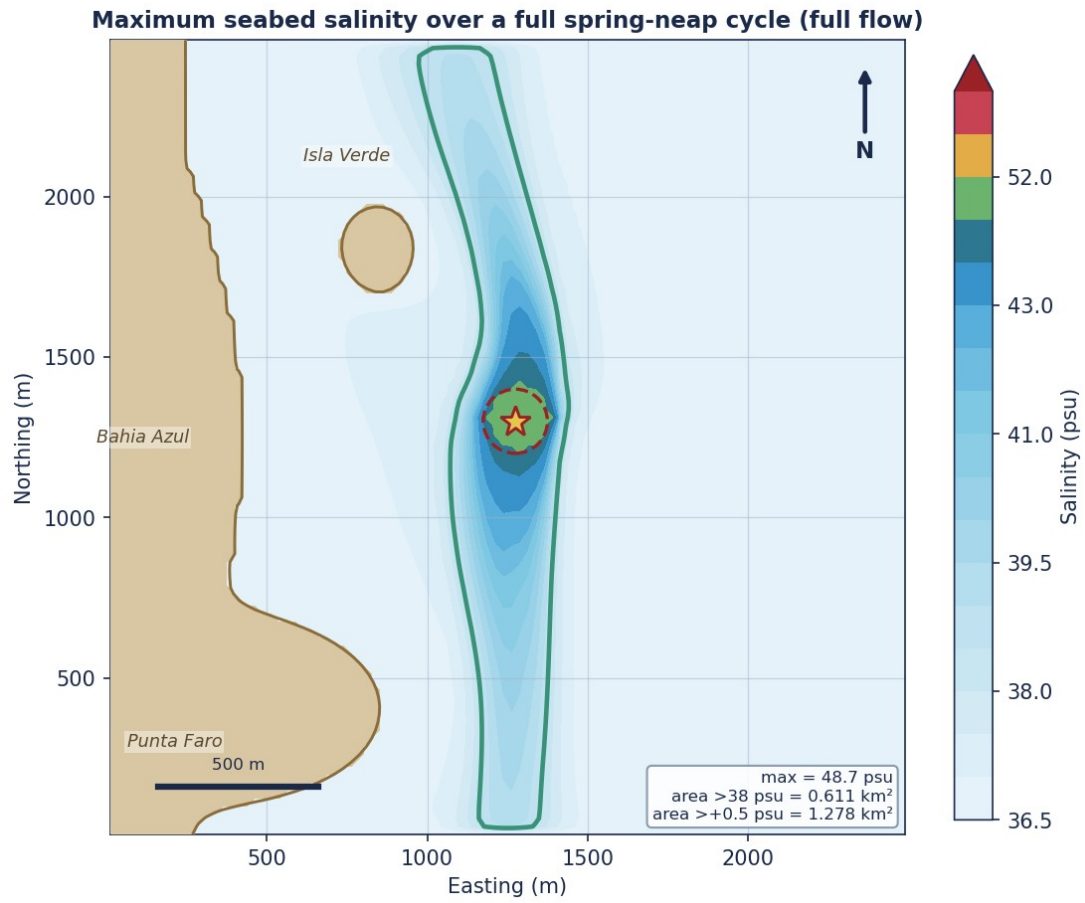


Figure. Maximum seabed salinity over a full spring-neap cycle (full flow)

6 Far-Field Dispersion

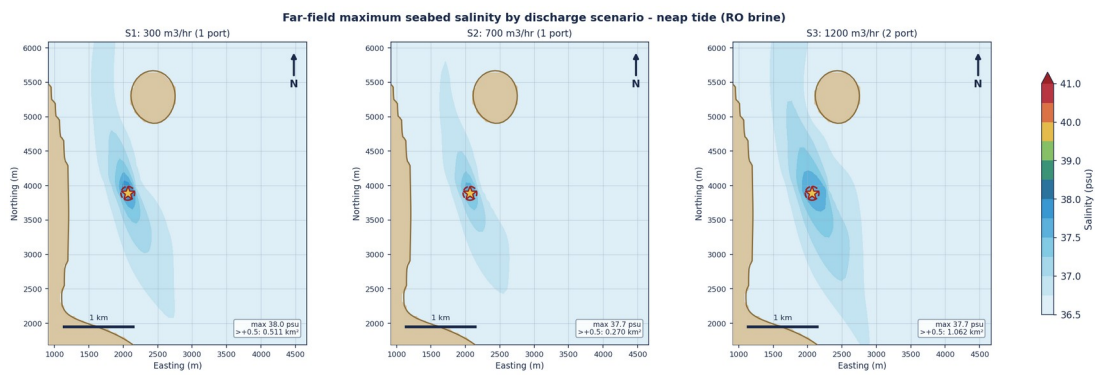


Figure. Far-field maximum seabed salinity by discharge scenario - neap tide (RO brine)

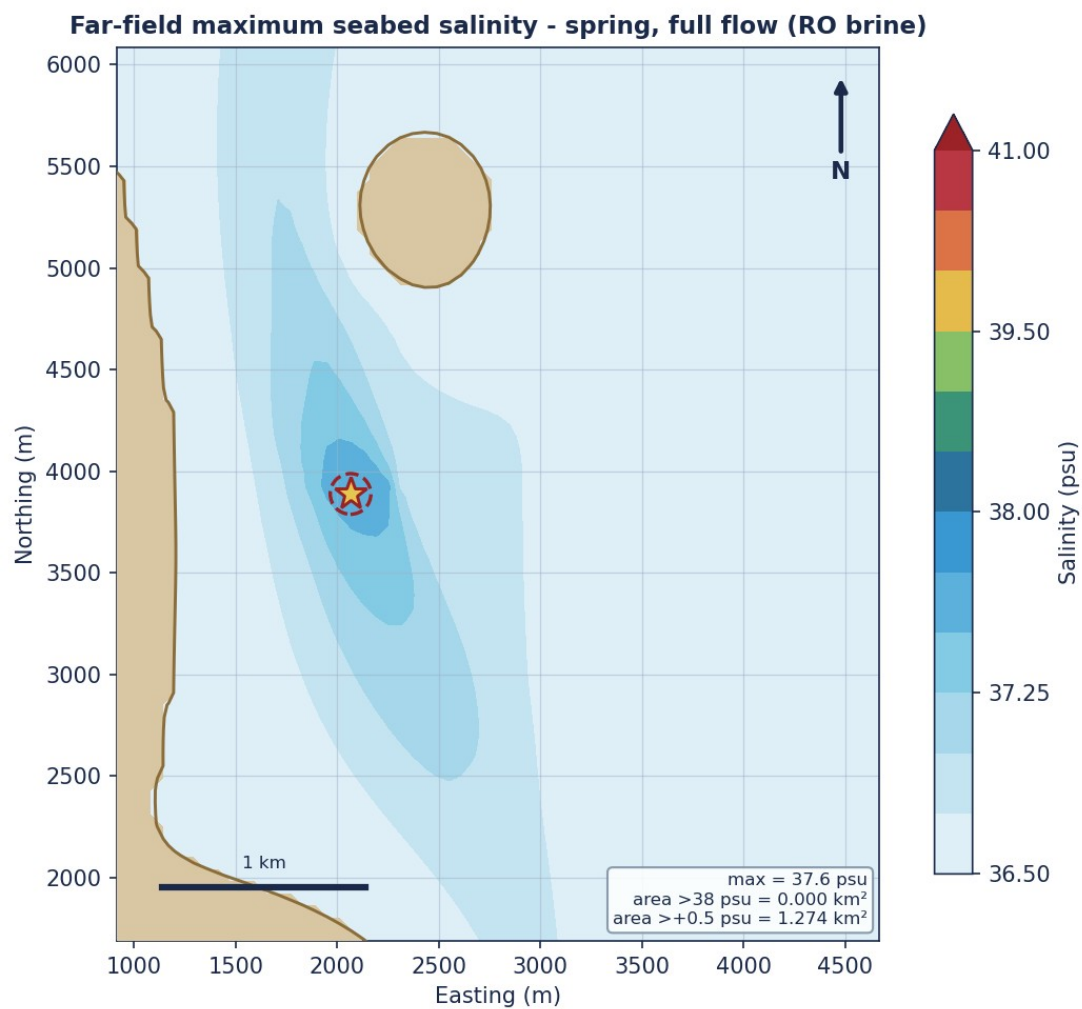


Figure. Far-field maximum seabed salinity - spring, full flow (RO brine)

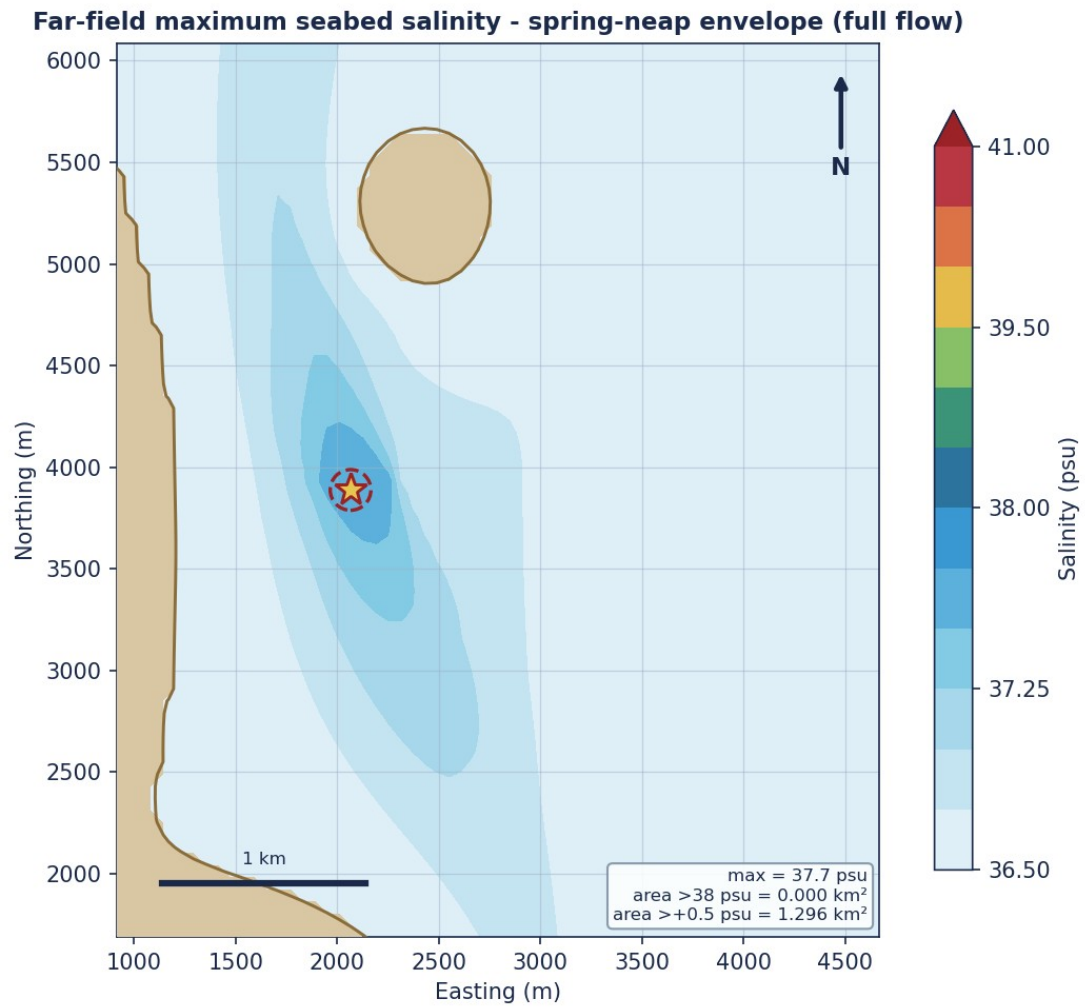


Figure. Far-field maximum seabed salinity - spring-neap envelope (full flow)

Far-field maximum seabed salinity - phase-2 saturated cavern brine (worst case)

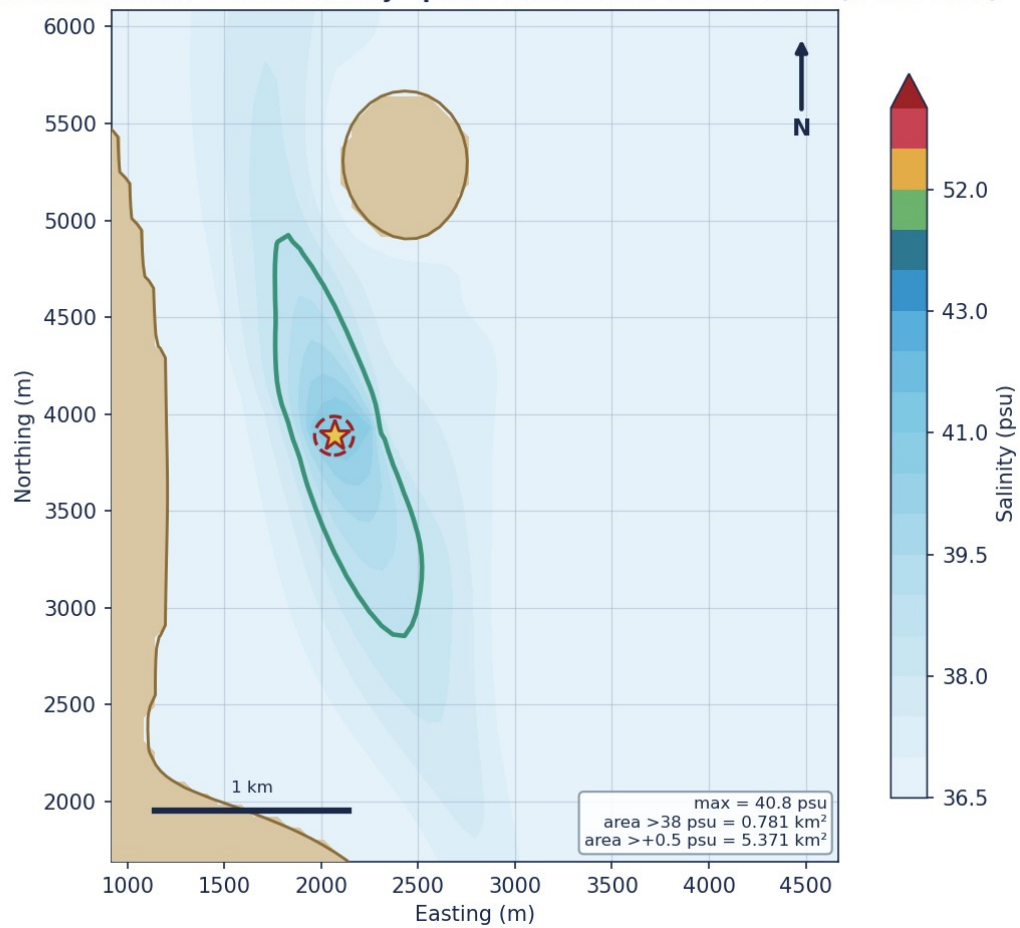


Figure. Far-field maximum seabed salinity - phase-2 saturated cavern brine (worst case)

Table. Salinity-increment influence areas and diffusion distances per scenario.

scenario	flow_m3hr	area_0p5 psu_km2	area_1p0 psu_km2	area_1p5 psu_km2	diff_dist_ parallel_ m	diff_dist_t ransverse_ _m	mixing_zo ne_radius_ _m	max_seab ed_salinit y_psu
S1	300	0.5112	0.0792	0.0	1805.0	348.0	0.0	37.97
S2	700	0.27	0.0252	0.0	1137.0	264.0	0.0	37.66
S3	1200	1.062	0.1548	0.0	2656.0	539.0	0.0	37.68

7 Diffuser Design Optimisation and Layout

The effect of diffuser port discharge angle on near-field dilution was assessed, and a recommended layout selected. Cross-shore salinity and current-speed transects and the marine-constraints overlay support the siting decision.

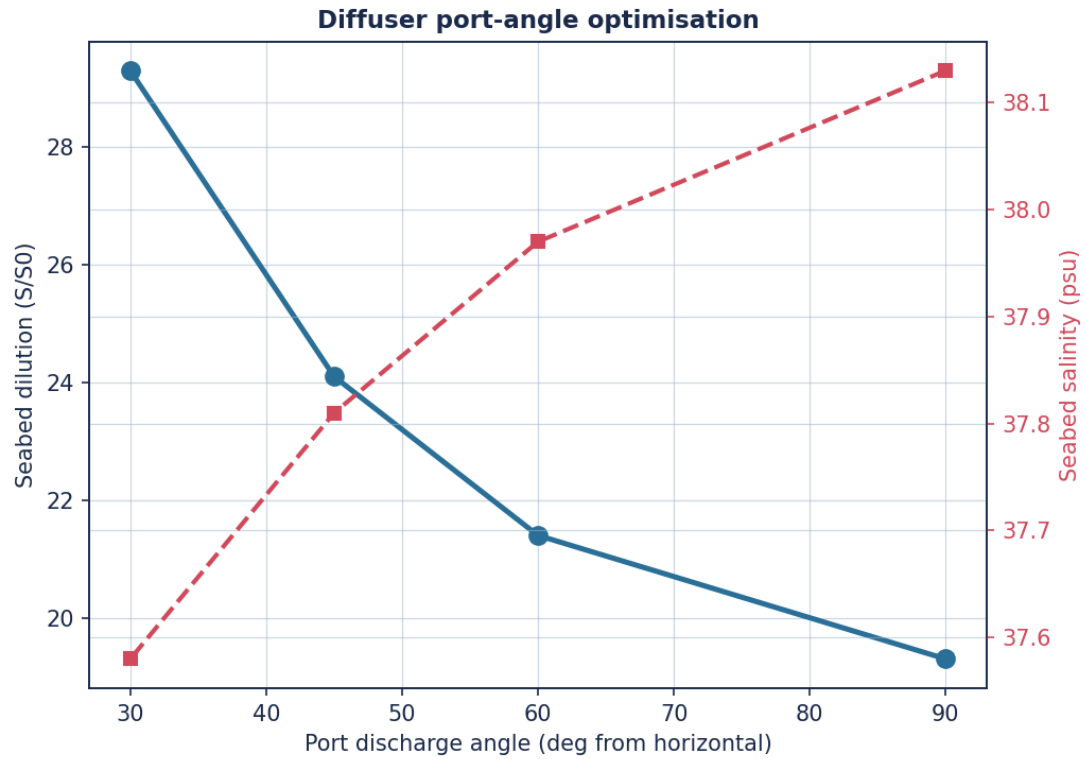


Figure. Diffuser port-angle optimisation: dilution and seabed salinity vs discharge angle.

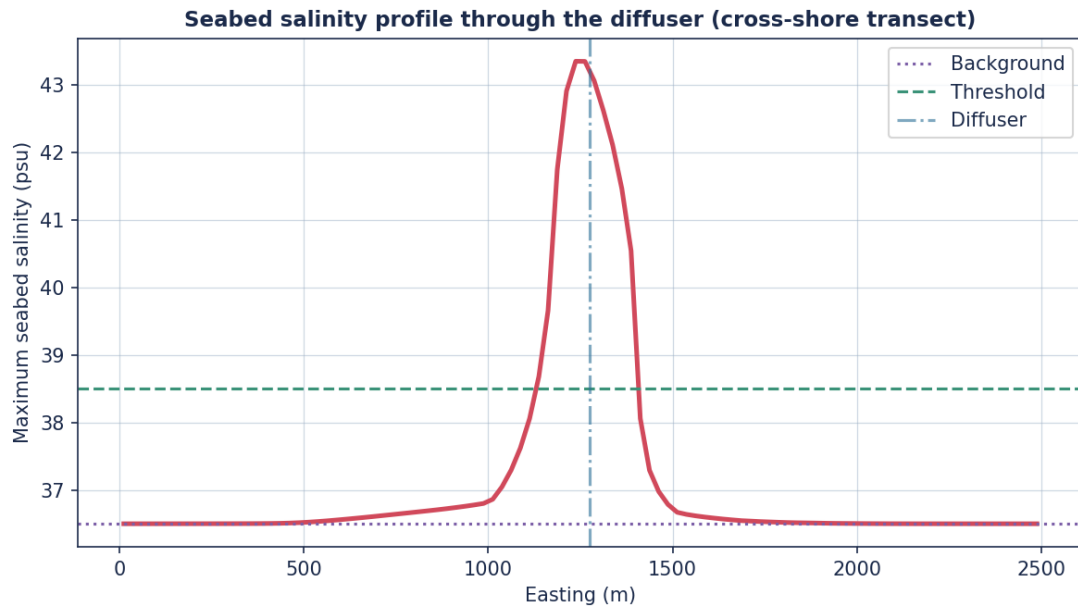


Figure. Maximum seabed salinity along a transect through the diffuser.

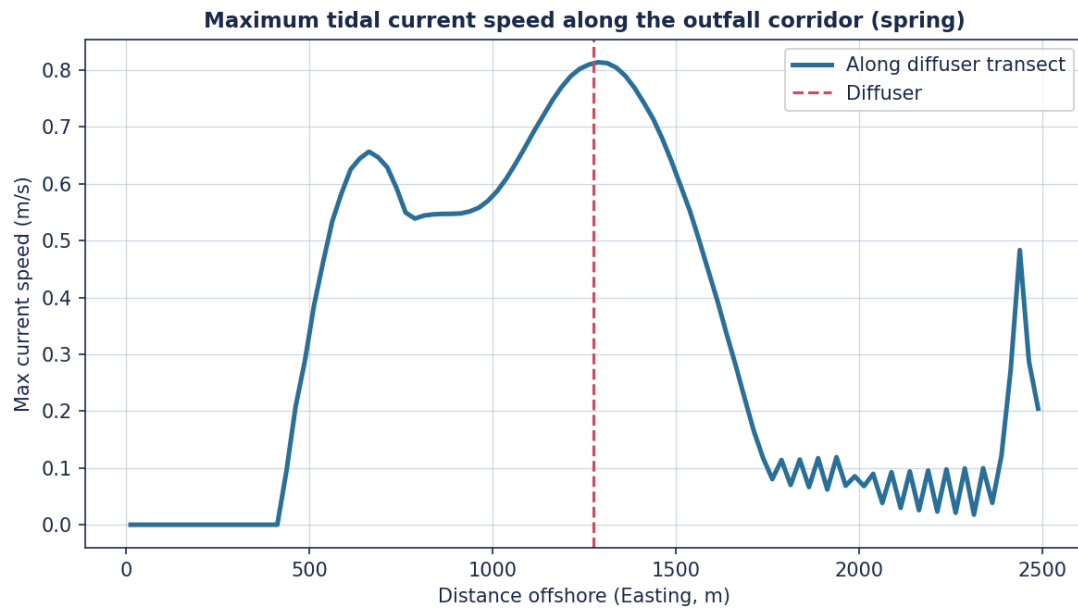


Figure. Maximum current speed along the outfall corridor (spring tide).

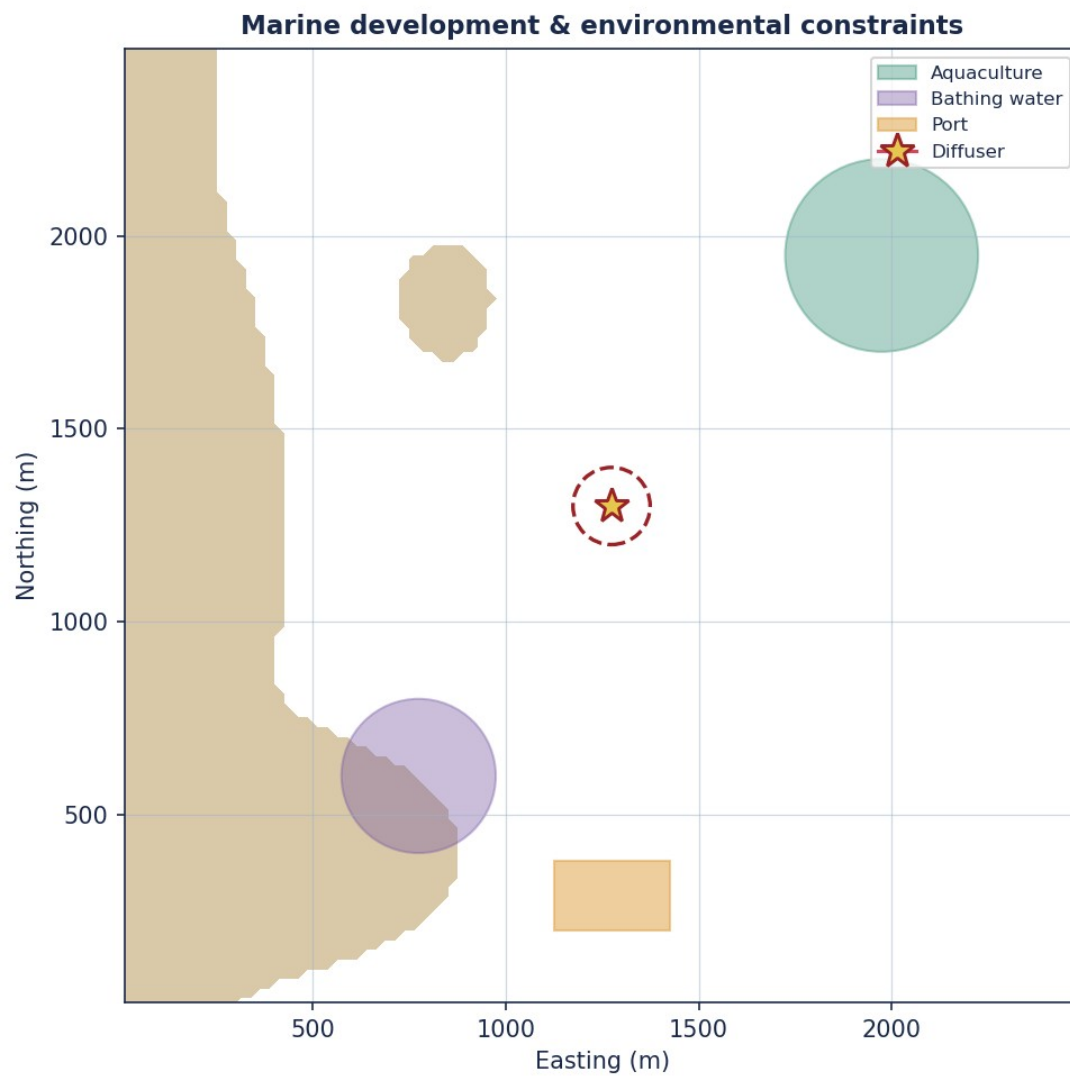


Figure. Distribution of marine development and environmental constraints relative to the mixing zone.

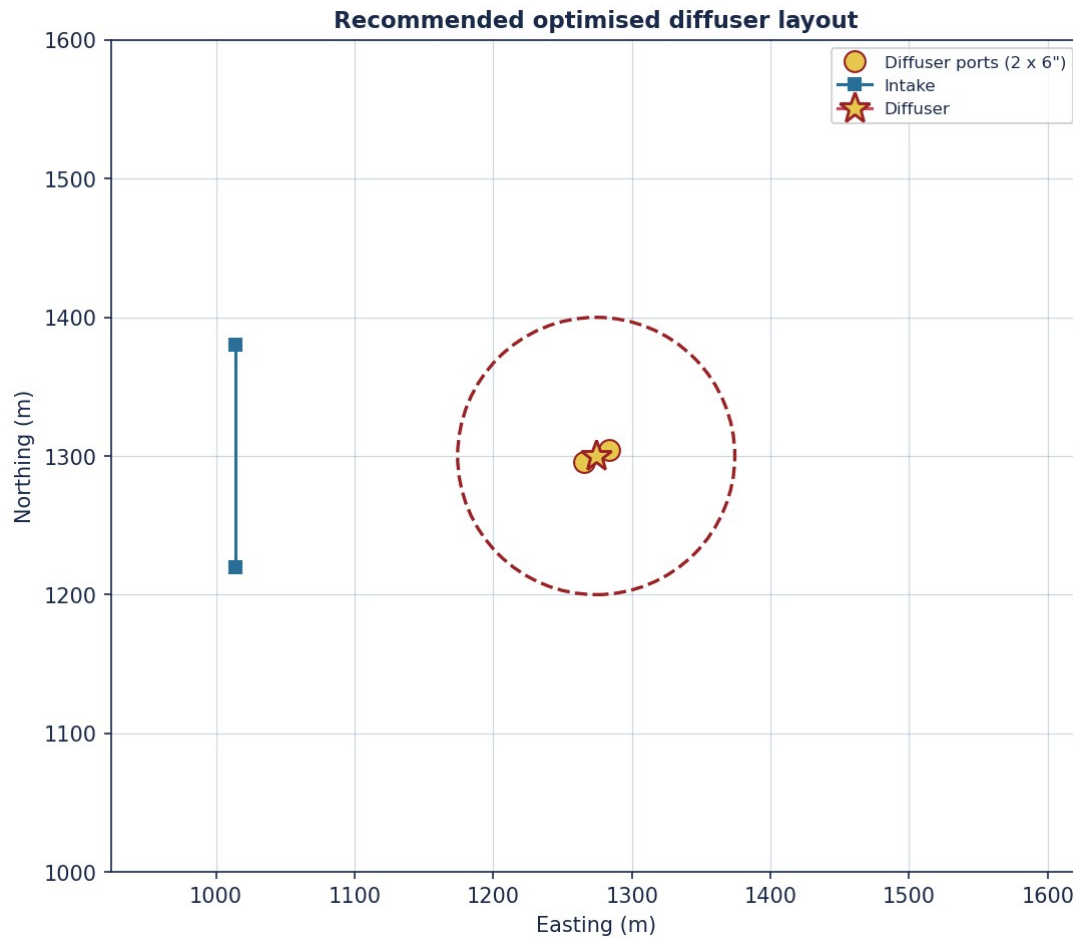


Figure. Recommended optimised diffuser/intake layout (inclined two-port diffuser).

8 Outfall Siting and Intake Recirculation

Following the desalination-siting methodology, the salinity rise at the plant intake was assessed as a function of outfall distance from shore under steady northward, southward and weak-southward currents. The recirculation criterion is a salinity rise at the intake of no more than 0.1 ppt above ambient. The weak southward current is the critical (least-flushing) condition.

Table. Rise in salinity at intake above ambient - southward current 0.25 m/s.

outfall_dist_m	rise_min_ppt	rise_max_ppt	rise_avg_ppt
600.0	0.004	0.068	0.023
800.0	0.001	0.011	0.003
1000.0	0.0	0.002	0.001
1200.0	0.0	0.0	0.0

Table. Rise in salinity at intake above ambient - weak southward current 0.05 m/s.

outfall_dist_m	rise_min_ppt	rise_max_ppt	rise_avg_ppt
----------------	--------------	--------------	--------------

600.0	0.016	0.061	0.035
800.0	0.007	0.024	0.014
1000.0	0.003	0.011	0.006
1200.0	0.002	0.005	0.003

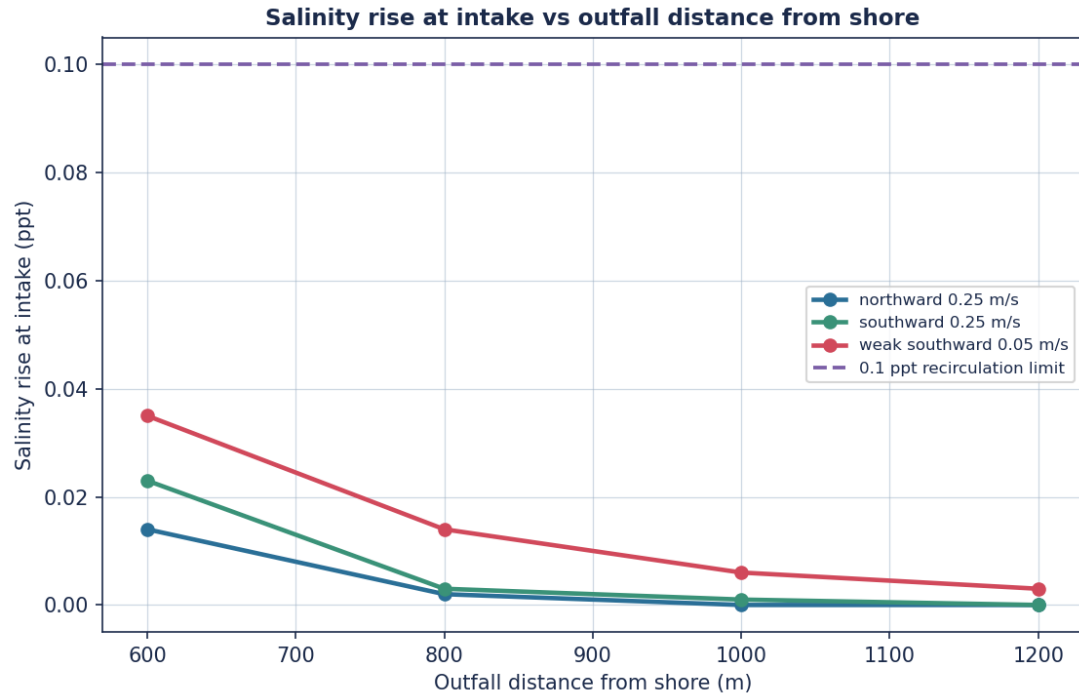


Figure. Variation of salinity rise at the intake with outfall distance from shore (recirculation criterion).

Contours of salinity rise - outfall 600 m from shore (weak southward current)

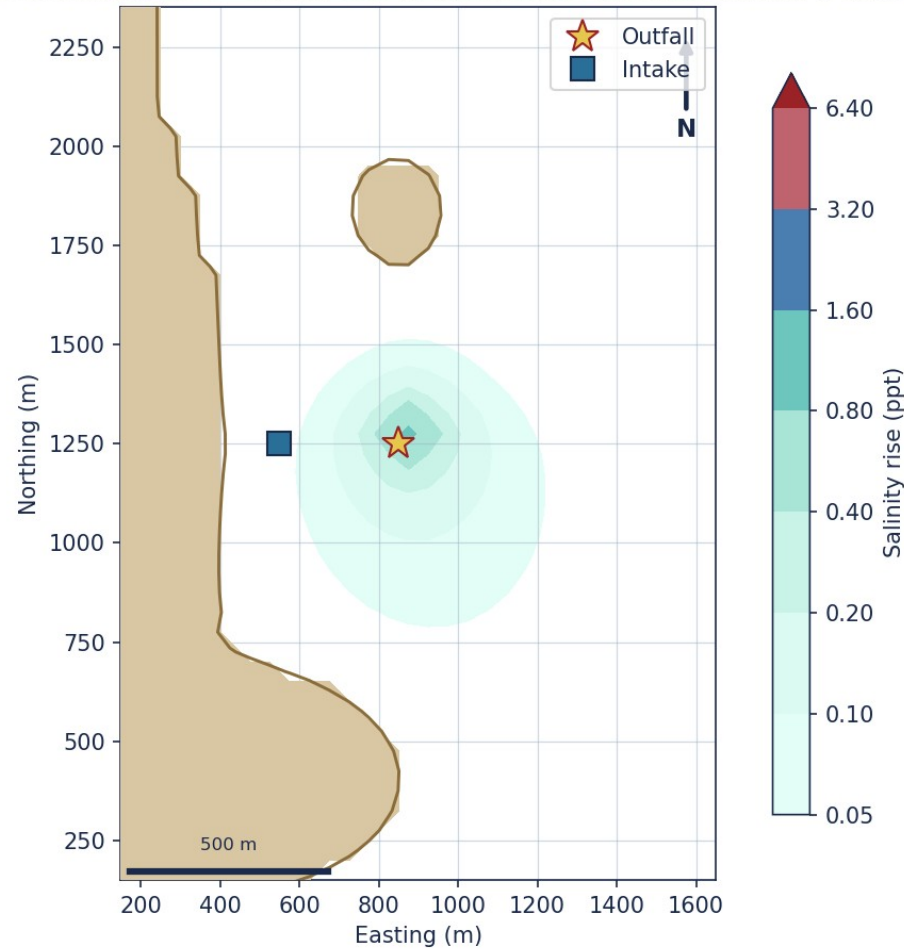


Figure. Salinity-rise contours, outfall 600 m from shore, weak southward current.

Contours of salinity rise - outfall 1200 m from shore (weak southward current)

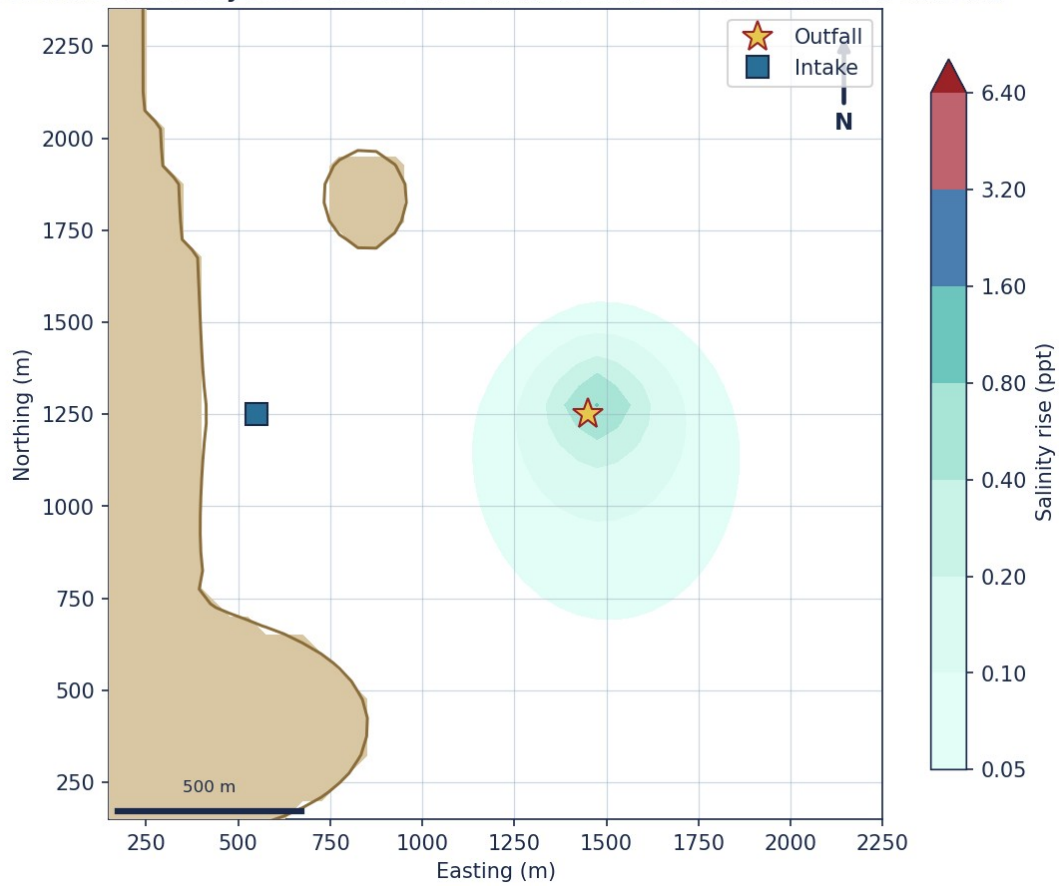


Figure. Salinity-rise contours, outfall 1200 m from shore, weak southward current.

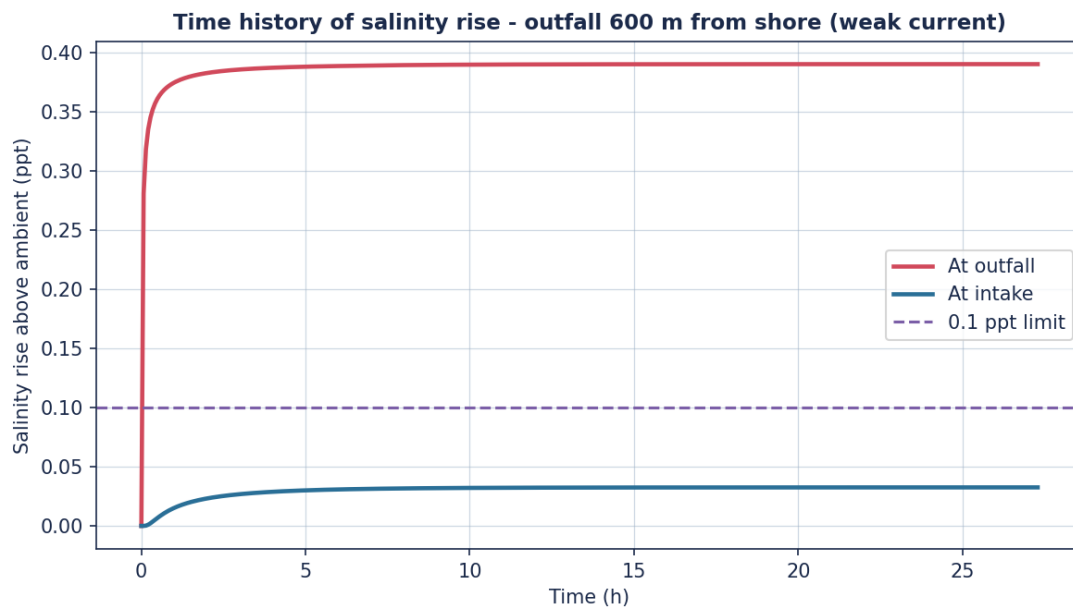


Figure. Time history of salinity rise at intake and outfall, outfall 600 m from shore.

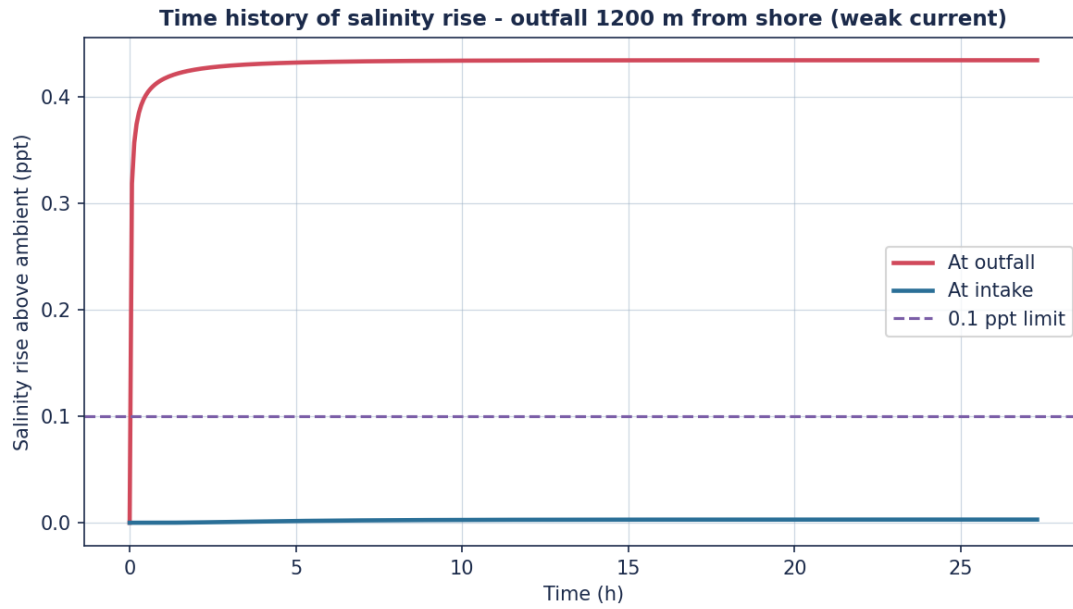


Figure. Time history of salinity rise at intake and outfall, outfall 1200 m from shore.

9 Other Impurities and Water-Quality Compliance

Table. Trace-impurity concentrations vs Environmental Quality Standards after dilution.

parameter	brine_ugL	seawater_u gL	eqs_ugL	dilution	predicted_u gL	margin_x	compliant
Arsenic	3.2	0.9	25.0	53.0	0.944	26.5	Yes
Boron	3450.0	4300.0	7000.0	53.0	4283.902	1.6	Yes
Cadmium	0.3	0.04	2.5	53.0	0.045	55.6	Yes
Chromium	1.1	0.8	15.0	53.0	0.806	18.6	Yes
Copper	1.8	1.1	5.0	53.0	1.113	4.5	Yes
Lead	2.0	0.1	25.0	53.0	0.136	183.8	Yes
Mercury	0.02	0.015	0.3	53.0	0.015	19.9	Yes
Nickel	2.9	0.3	30.0	53.0	0.349	85.9	Yes
Zinc	4.0	1.2	40.0	53.0	1.253	31.9	Yes

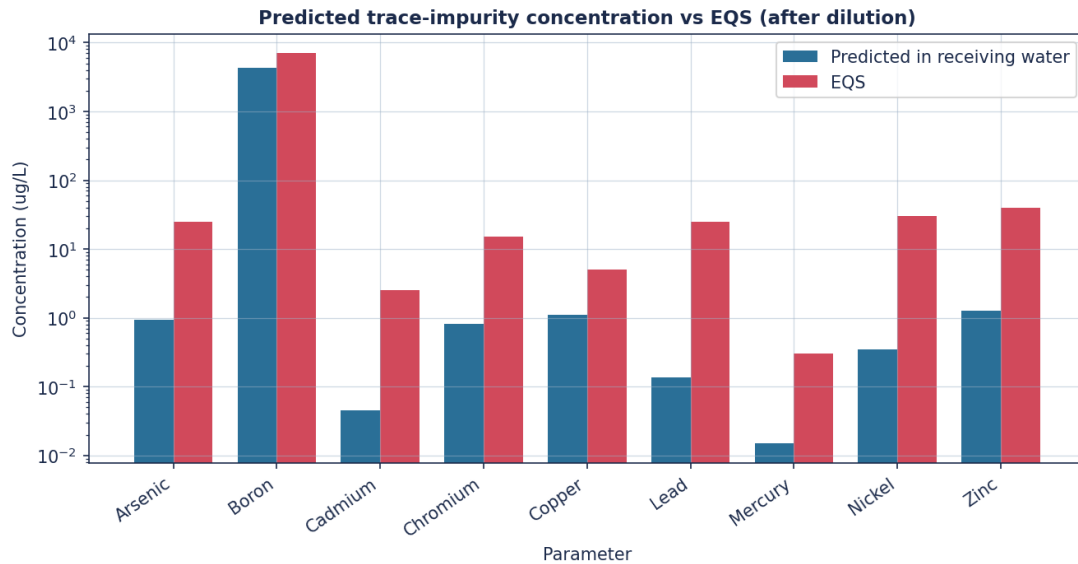


Figure. Predicted receiving-water trace-impurity concentrations vs EQS thresholds.

10 Validation and Data Sources

UBDS was validated against the following independent benchmarks:

- Equation of state: maximum error 0.402% vs reference seawater/brine densities (UNESCO EOS-80; CRC Handbook).
- Momentum jet: dilution slope 0.228 per z/D and spread 0.114 vs canonical 0.25-0.32 and ~ 0.11 (Fischer, List, Koh, Imberger & Brooks, 1979).
- Pure plume: volume-flux exponent 1.596 vs the $5/3$ power law (Morton, Taylor & Turner, 1956).
- Inclined 60 deg dense jet: $z_t/(D.Fr) = 1.534$ and $S_r/Fr = 0.891$ vs published ranges 1.6-2.2 and 0.4-0.6 (Papakonstantis, Christodoulou & Papanicolaou, 2011).

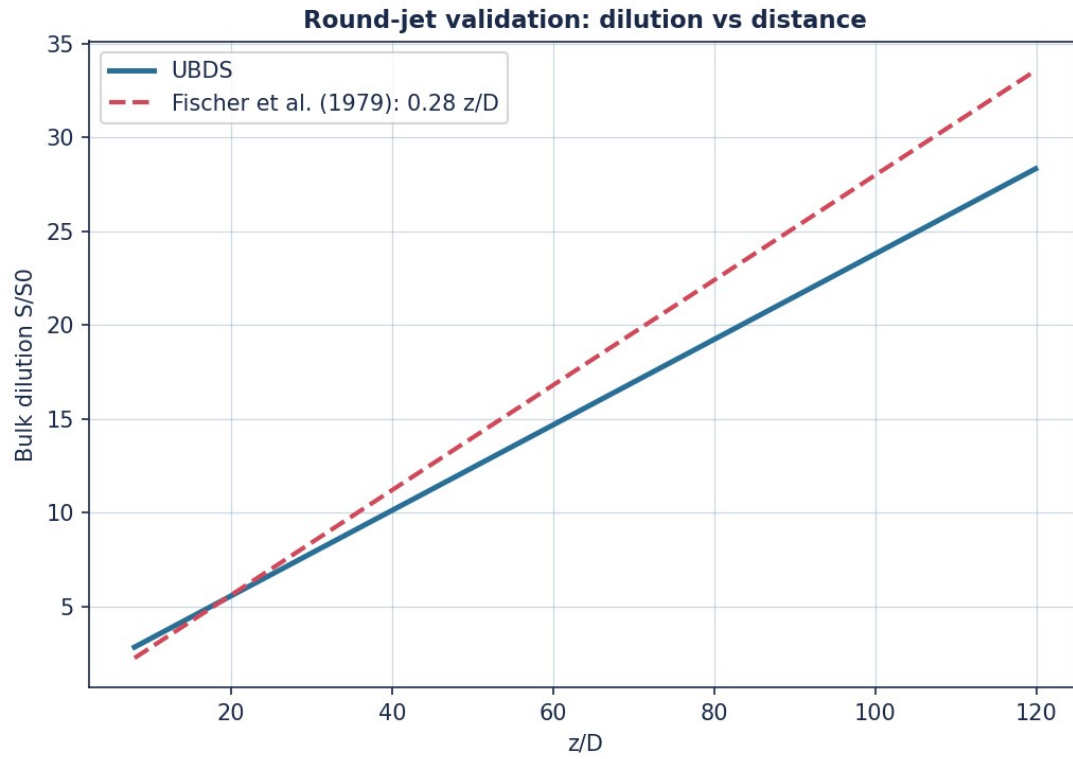


Figure. Momentum-jet dilution: UBDS vs the canonical self-similar law (Fischer et al. 1979).

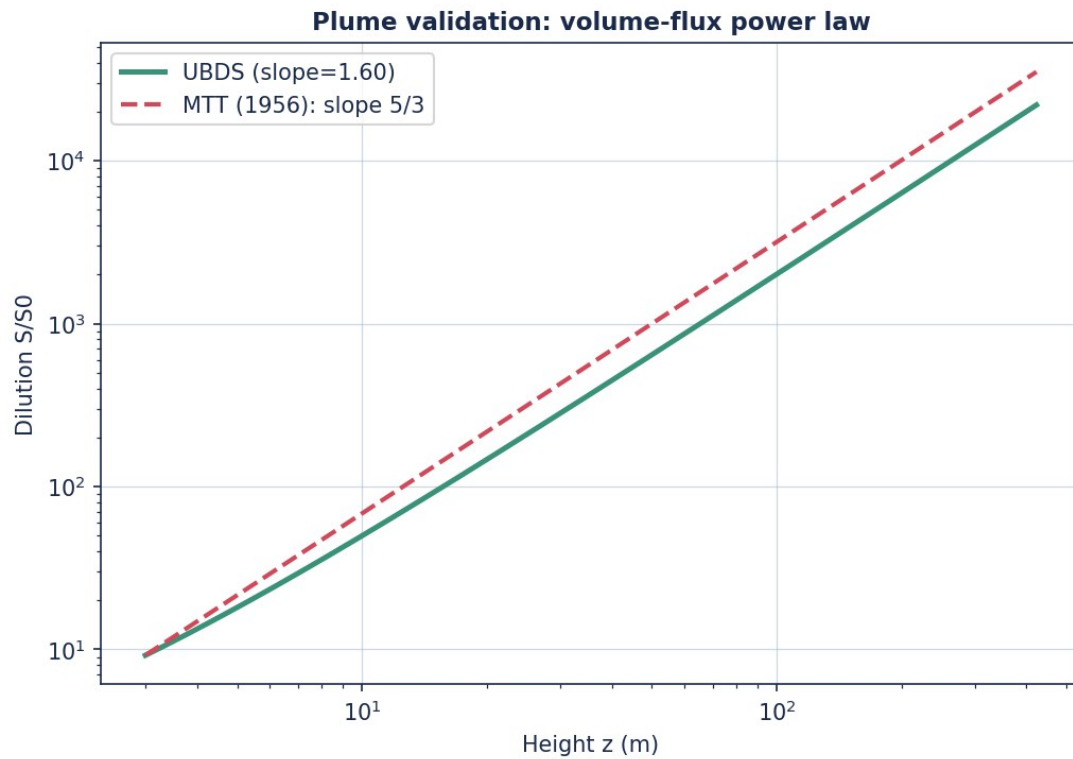


Figure. Pure-plume dilution: UBDS vs the $5/3$ power law (Morton, Taylor & Turner 1956).

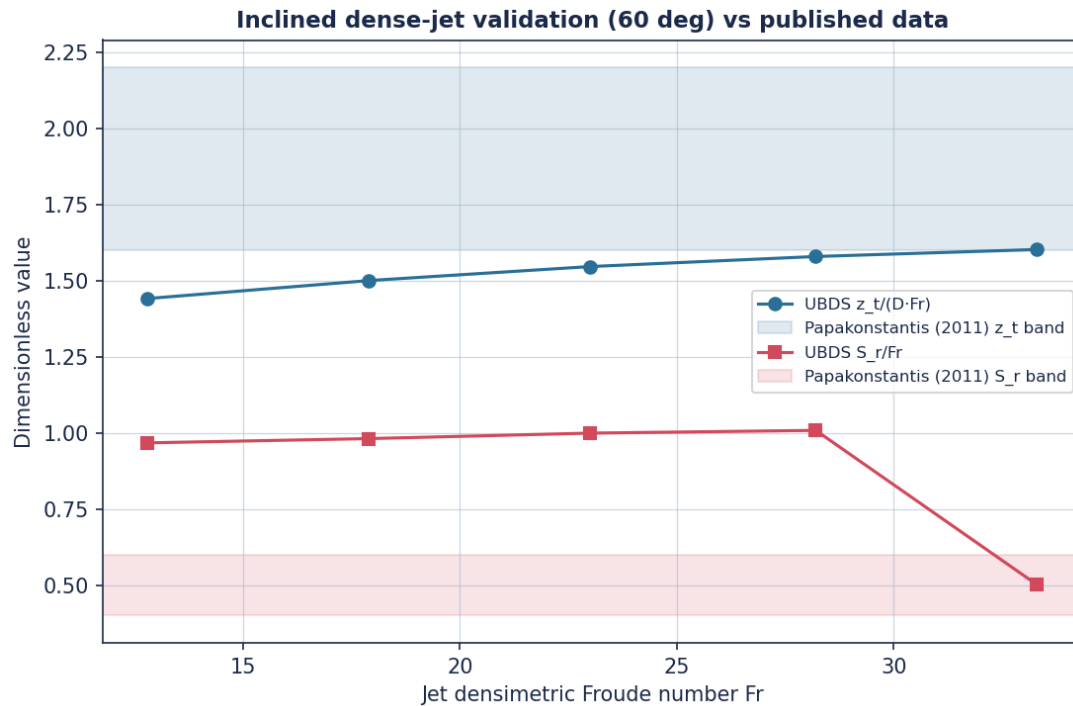


Figure. Inclined 60 deg dense-jet dimensionless rise and dilution vs published experimental bands.

Table. Equation-of-state validation.

salinity_psu	temp_C	ref_density	model_density	error_pct	source
0.0	25.0	997.05	997.05	-0.0	Pure water (CRC)
35.0	15.0	1025.97	1025.97	0.0	EOS-80 (Millero & Poisson 1981)
35.0	25.0	1023.34	1023.34	0.0	EOS-80
38.0	20.0	1026.95	1027.05	0.01	EOS-80
260.0	20.0	1200.0	1204.82	0.402	Saturated NaCl brine (CRC Handbook)

Table. Inclined dense-jet dimensionless validation.

Fr	zt_over_DFr	xr_over_DFr	Sr_over_Fr
12.8	1.441	2.449	0.967
17.9	1.5	2.538	0.981
23.0	1.546	2.597	0.999
28.2	1.579	2.627	1.008
33.3	1.602	1.472	0.501

9.1 Data sources

- Fischer, H.B., List, E.J., Koh, R.C.Y., Imberger, J. & Brooks, N.H. (1979). Mixing in Inland and Coastal Waters. Academic Press. [jet/plume dilution laws]

- Morton, B.R., Taylor, G.I. & Turner, J.S. (1956). Turbulent gravitational convection from maintained and instantaneous sources. Proc. R. Soc. Lond. A 234, 1-23. [plume 5/3 law]
- Papakonstantis, I.G., Christodoulou, G.C. & Papanicolaou, P.N. (2011). Inclined negatively buoyant jets 1 & 2. J. Hydraulic Research 49(1), 3-22. [inclined dense-jet data]
- Roberts, P.J.W., Ferrier, A. & Daviero, G. (1997). Mixing in inclined dense jets. J. Hydraulic Engineering 123(8), 693-699. [dense-jet dilution]
- Millero, F.J. & Poisson, A. (1981). International one-atmosphere equation of state of seawater. Deep-Sea Research 28A, 625-629; UNESCO (1981) Tech. Papers Mar. Sci. 36. [EOS-80]
- Jirka, G.H. (2004). Integral model for turbulent buoyant jets in unbounded stratified flows. Part 1. Environmental Fluid Mechanics 4, 1-56. [integral jet model]
- Lee, J.H.W. & Chu, V.H. (2003). Turbulent Jets and Plumes - A Lagrangian Approach. Kluwer. [entrainment closure]
- CRC Handbook of Chemistry and Physics. [saturated NaCl brine density]
- Synthetic site-specific data (bathymetry, tides, ADCP currents, CTD salinity/temperature, brine chemistry) generated by the author for this case study; all values are documented and engineering-credible, and none are taken from any real consent.

11 Conclusions

The UBDS solver provides a single, open, physically-validated framework spanning the near-, medium- and far-field dispersion of brine and other buoyant effluents. Applied to the Bahia Azul case study with fully documented, credible site-specific inputs, it demonstrates that the proposed inclined two-port diffuser achieves effective initial dilution and confines significant salinity increments and all trace impurities to within the regulatory mixing zone, across the full range of discharge flows and tidal states.

12 Animated Outputs

The following animated outputs (GIF files in outputs/animations/) show the time-evolving brine dispersion and tidal current field over a tidal cycle. A static filmstrip of the dispersion animation is embedded below; the full animations are in outputs/animations/.

- ANIM_brine_dispersion_spring.gif - Brine dispersion over a tidal cycle (full flow, spring)
- ANIM_brine_dispersion_neap.gif - Brine dispersion over a tidal cycle (full flow, neap)
- ANIM_current_field_spring.gif - Tidal current field over a cycle (spring)

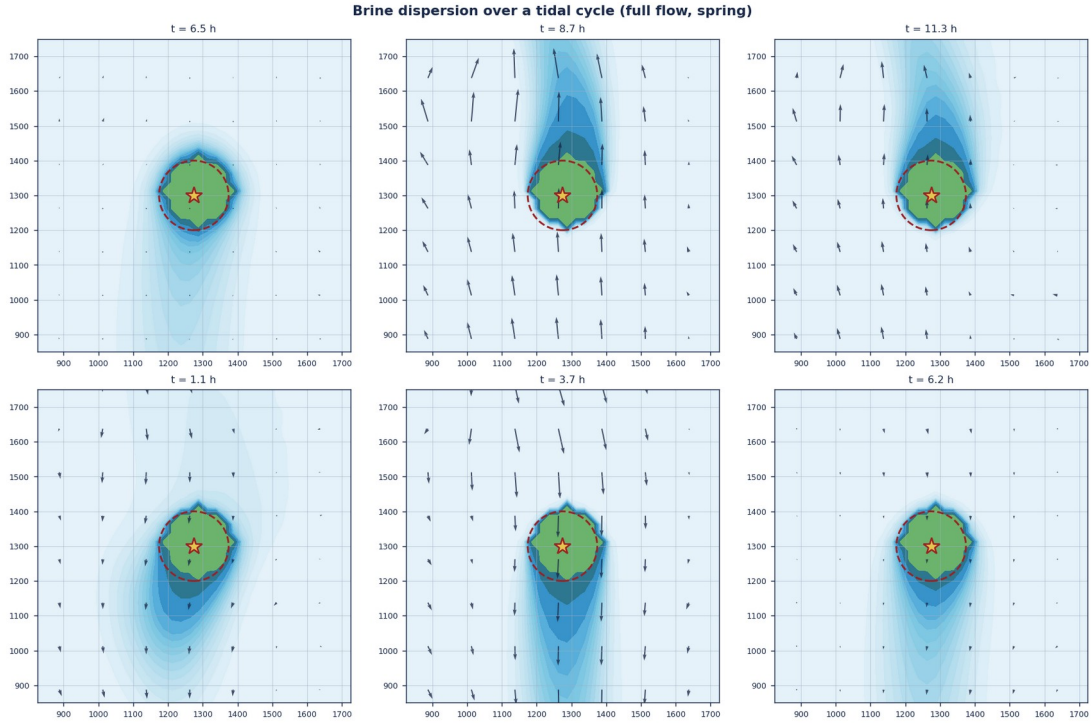


Figure. Brine dispersion over a tidal cycle (full flow, spring) (filmstrip of the animation).

Appendix A Representative Output Figures from the Reference Studies

For methodological comparison, representative output figures from the three reference studies that guided this work are reproduced below (author's reference materials, attributed). They illustrate the output categories - mesh/bathymetry, current calibration, near-field trajectories/dilution, salinity envelopes and snapshots, and intake/outfall time histories - that the UBDS outputs reproduce for the Bahia Azul case.

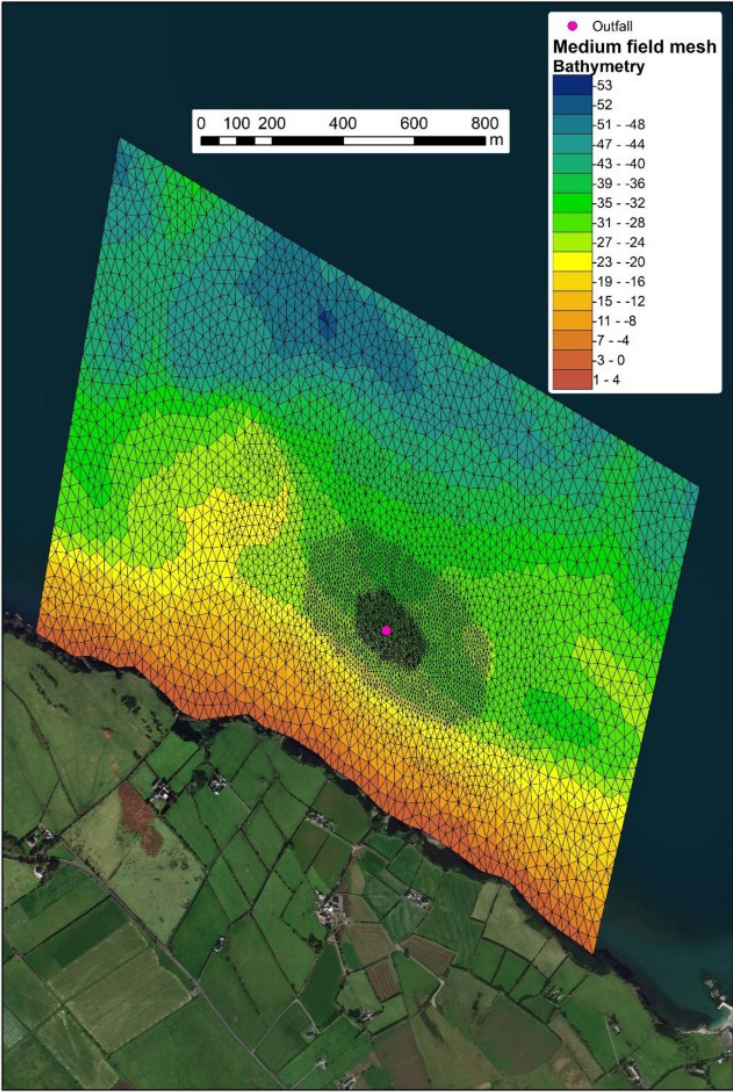


Figure 1-1 – Mesh and Bathymetry of the Islandmagee Medium Field Tidal Model

Figure. Reference Study 1 (salt-cavern brine outfall): model mesh, bathymetry and current-meter comparison.

3 INITIAL DILUTION MODELLING

The initial dilution study examines the dispersion of the brine discharged from the outfall diffuser jets in the immediate area around the diffuser. For this study the initial dilution achieved by the brine discharge from the proposed outfall was modelled using the US EPA "Plumes" software utilising the UM3 routine. The Visual Plumes programme simulates the flow of the outlet jets under the influence of density, temperature and velocity. The programme is primarily a near-field model and a mid- and far-field model is therefore also required to simulate the plume dispersion over the wider area.

UM3 is an acronym for the three-dimensional Updated Merge (UM) model. UM3 simulates single and multi-port submerged discharges. UM3 is a Lagrangian model that assumes that the plume is in a steady state. However, ambient and discharge conditions can change as long as they do so over time scales which are long compared to the time in which a discharged element reaches the end of the initial dilution phase. To make UM three-dimensional, the model includes an entrainment term corresponding to the third-dimension i.e. a cross-current term. As a result, single-port plumes are simulated as truly three-dimensional entities. Merged plumes are simulated by distributing the cross-current entrainment over all plumes. Dilution from diffusers oriented parallel to the current is estimated by limiting the effective spacing to correspond to a cross-diffuser flow angle of 20 degrees.

3.1 Initial Dilution Model Results

The initial dilution modelling was undertaken using a diffuser with either 1 or 2 ports depending on the flow range as stated in Section 2.2. The diffuser ports were at 20m centres with the port exits discharging vertically upwards at approximately 0.75m above the seabed. Following discharge it is expected that the brine will sink down towards the seabed due to the density of the brine solution, with the initial trajectory depending on the tidal velocity. Consequently the UM3 model was run for a variety of spring and neap flow conditions to simulate the initial dilution achieved at various stages of the tidal cycle.

3.1.1 Scenario 1: 250m³/hr via a Single Port

This scenario represents the early stages of project development when the leaching of the caverns is just commencing and the full flow of leaching water is not required. For this stage of the process it was assumed that one port on the diffuser would be blanked off and hence the 250m³/h would be discharged via one diffuser port.

The trajectories for the discharge from a 6" Tideflex port across the range of tidal velocities are presented in Figure 3-1. The trajectory in red relates to low tidal velocity of <0.1m/s which would be experienced during the turning tide, the trajectory in blue relates to tidal velocity of 0.3 m/s which would represent peak velocities during neap tide. The trajectory in green relates to a typical spring tidal velocity of 0.6 m/s.

Table 4-1: Table showing Water Column division into Layers within the Models in terms of % of overall water depth assigned to each layer.

	14 Layer	9 Layer	4 Layer
Layer 14	40	-	-
Layer 13	20	-	-
Layer 12	11	-	-
Layer 11	8.5	-	-
Layer 10	6.5	-	-
Layer 9	4.5	40	-
Layer 8	3	25	-
Layer 7	2	15	-
Layer 6	1.5	9.3	-
Layer 5	1	4.5	-
Layer 4	0.8	2.5	35
Layer 3	0.6	1.5	35
Layer 2	0.5	1.2	15
Bottom Layer 1	0.1	1	15

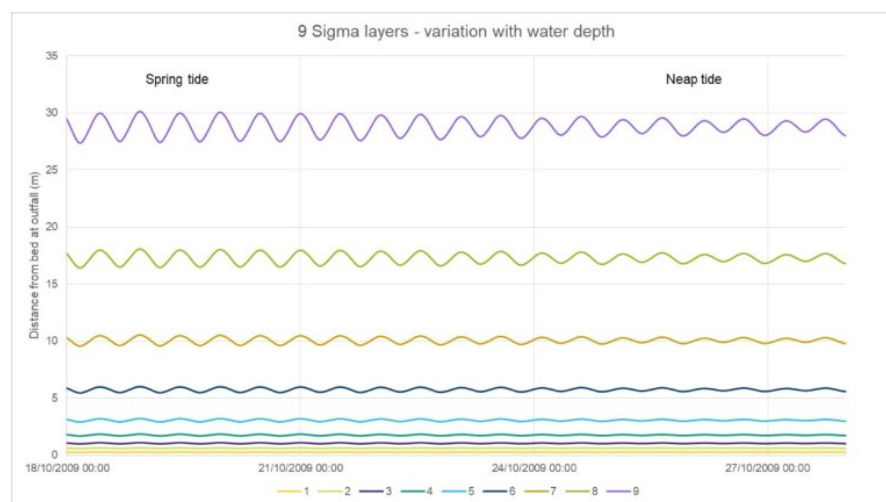


Figure 4-1: Variation of sigma layers with water depth (9 layers)

For the sensitivity testing the 1,000m³/hr scenario was selected as this represents the maximum input of salt and hence density effects will be more prevalent than with lower salt inputs. The results of the sensitivity testing in terms of maximum seabed salinity levels are presented in Figure 4-2 to Figure 4-4 for the 4 layer, 9 layer and 14 layer models respectively. It should be noted in all figures presented the colour contour bands are not linear with smaller intervals at the lower salinity values in order to fully illustrate the range

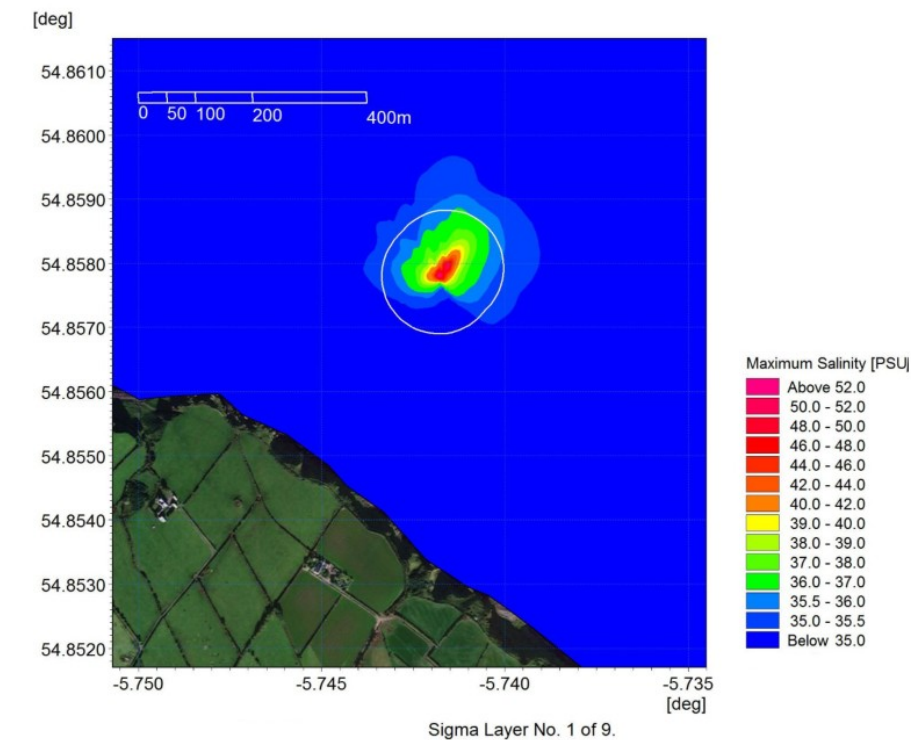


Figure 4-8: Maximum Salinity at Bed – Neap Tide

Figure 5-4 respectively. In all figures presented the colour contour bands are not linear with smaller intervals used at the lower salinity and the 36 psu threshold is shown in yellow.

The salinity increases predicted beyond the initial mixing zone in all cases are relatively small; the contour intervals plotted representing very small increases of 0.25-0.50 psu. A maximum far field increase in salinity of less than 4 psu i.e. total salinity <38 psu, was recorded throughout all the simulations which is commensurate with the levels predicted for the concept design considered at planning stage.

The baseline salinity data obtained from AFBI and NIEA at the initial project development stage indicated that background salinities in this part of the North Channel off Islandmagee can range between circa 30.5 psu and 34.8 psu. Even with an assumed background salinity of 34.2 any salinity increase in excess of the range normally experienced in seasonal variations (34.8 psu) is expected to be restricted to an area less than 300 metres from the outfall.

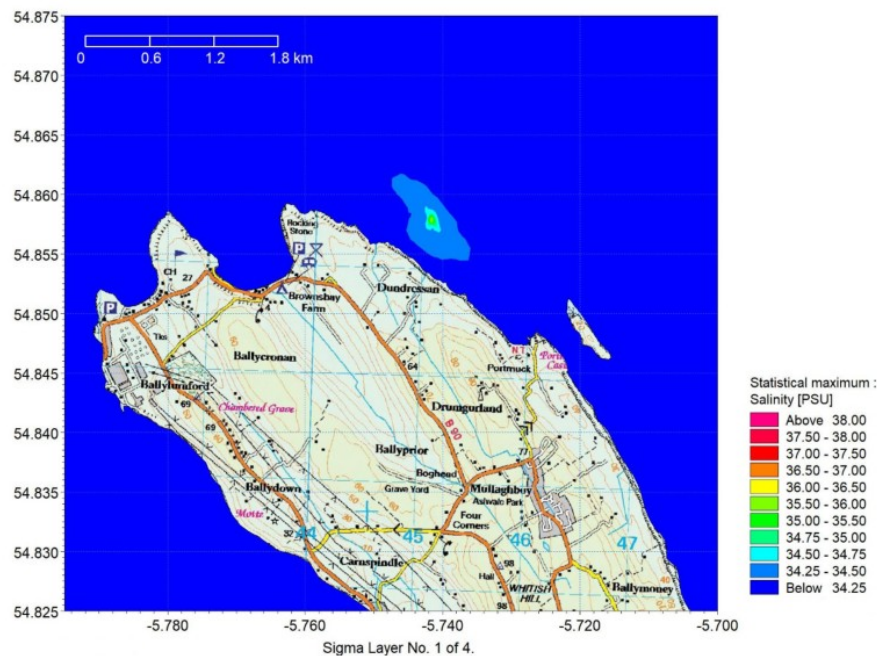


Figure 5-2: Maximum Seabed Salinity during a Neap Tidal Cycle – 250m³/hour Discharge

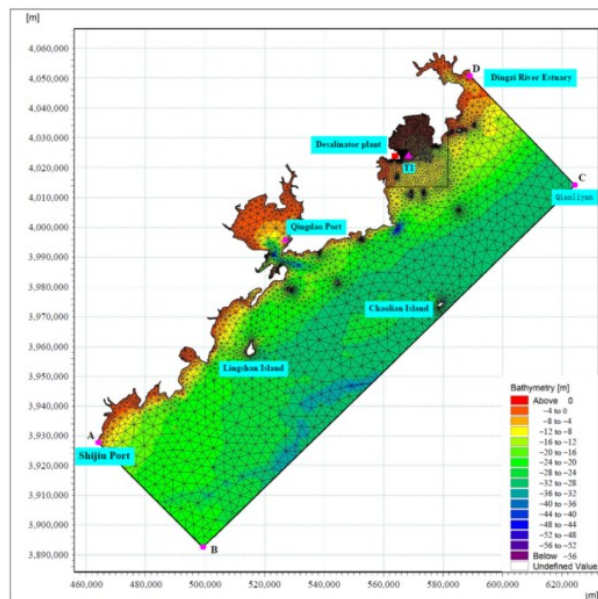


Figure 2. Grid distribution in the computational domain, and color-shaded bathymetry. A, B, C, and D are control points for open water boundaries. T1 is a tidal level monitoring point.

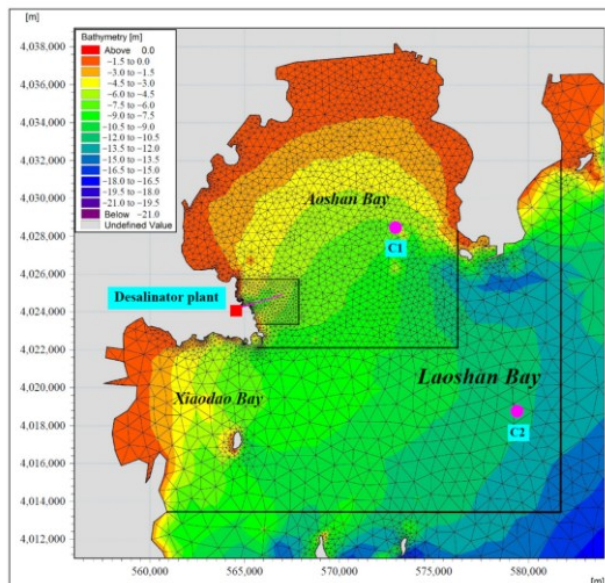


Figure 3. Local grid refinement in the engineered area, and locations of verification points. C1 and C2 are current monitoring points.

Figure. Reference Study 2 (Laoshan Bay desalination): computational mesh and bathymetry of the bay.

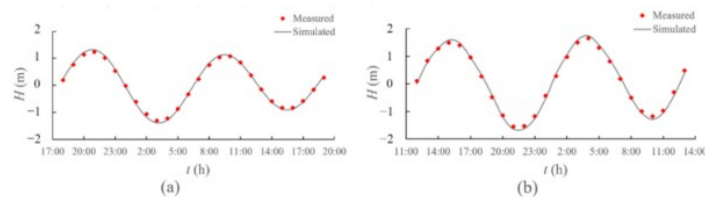


Figure 4. Comparison of the simulated and measured values of the tidal level at the Qingdao Port (a) and Station T1 (b).

The tidal flow field simulation results were verified by continuous 25 h measured current data and model calculation results from two stations in Laoshan Bay in April 2018. The positions of the tidal flow verification points are shown in Figure 3, and the tidal flow verification curves of the two stations are shown in Figures 5 and 6, respectively.

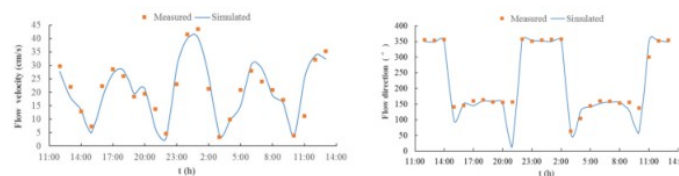


Figure 5. Comparison between the simulated and measured values of the flow velocity and direction at Station C1.

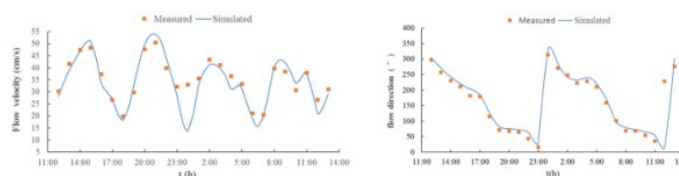


Figure 6. Comparison between the simulated and measured values of the flow velocity and direction at Station C2.

The calculated results of the model were found to be in good agreement with the measured tidal level data and ocean current data, properly reflecting the tidal level and ocean current characteristics in Laoshan Bay and its surrounding waters and meeting the simulation accuracy requirements of the hydrodynamic model and advection-diffusion model in this paper.

4. Results and Discussion

4.1. Simulation Results of Tidal Flow Field in Laoshan Bay

The simulation results of the tidal flow field in Laoshan Bay calculated using the model are shown in Figure 7. During the flood tide, the flow velocity in Laoshan Bay is generally 30–55 cm/s, and outside the bay the flow velocity is 60–70 cm/s. The flood current flows from NE to SW from the eastern sea area of Tianheng Town, then turns to NW after passing the headland and flows into Aoshan Bay on the north side. The flow velocity of the flood current in Aoshan Bay is typically 10–42 cm/s, and the flow velocity on the shore is relatively small, generally less than 10 cm/s. Near Nvdiao Island, the flow velocity of the flood current is relatively high, between 54 cm/s and 78 cm/s. The flood current in the area where the project is located flows from NNW to SSE, with a velocity between

10 cm/s and 36 cm/s. After passing the project area, the flood current along the coast passes around the Aoshantou headland and then turns west to Xiaodao Bay. The flow velocity in the Aoshantou sea area is relatively high, with a maximum of 72 cm/s. The flood current in the Xiaodao Bay generally flows from NE to SW at a velocity of 10–42 cm/s, while the velocity near the shore is low, generally less than 10 cm/s. Then the tidal current flows from N to S along the shore to the outside of the bay, and the velocity increases accordingly.

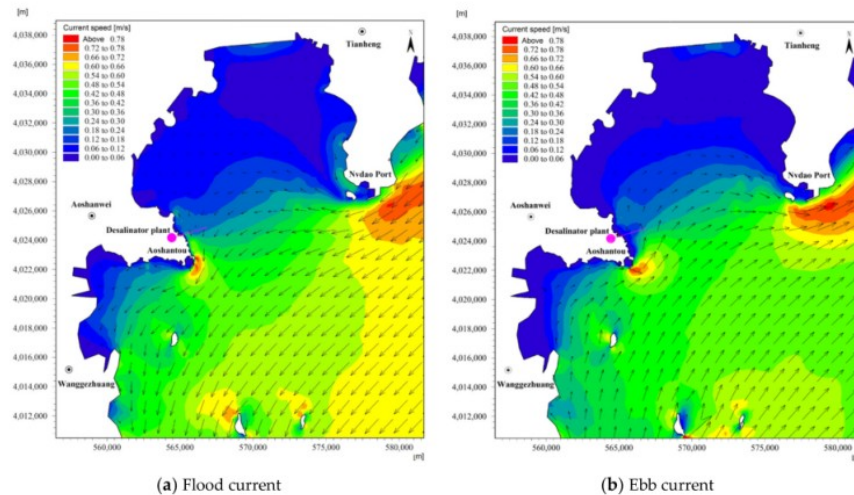


Figure 7. Hydrodynamic flow field simulation results.

During the ebb tide, the velocity in Laoshan Bay is generally 25–50 cm/s, and the velocity outside the bay is 55–65 cm/s. The northern ebb current flows from NW to SE, bypasses Nvdiao Island, then turns to the NE and flows to the sea on the east side of Tianheng Town. The flow velocity in Aoshan Bay is 10–48 cm/s, and the velocity on the shore is low, generally less than 10 cm/s. The ebb tide velocity on the south side of Nvdiao Island is relatively high, between 54 cm/s and 72 cm/s. Along the south side of the sea, the ebb tide flows from S to N outside the bay, and then from SW to NE inside Xiaodao Bay, and the flow velocity decreases accordingly. The flow velocity of the ebb tide in Xiaodao Bay is 10–36 cm/s, and the flow velocity on the shore is relatively low, generally less than 10 cm/s. After bypassing Aoshantou Headland, the ebb tidal current passes through the sea area of the project area, flowing from SSE to NNW at a flow velocity of 10–30 cm/s. The sea area south of Aoshantou has a high flow velocity of up to 68 cm/s.

The calculation results of the tidal flow field and the conclusions of historical research results are consistent [22,23], and the numerical simulation results properly reflect the distribution law of the tidal flow field in Laoshan Bay.

4.2. Simulation Results of Salinity Field after Brine Discharge

The diffusion process of brine in Laoshan Bay is mainly affected by tidal currents. The calculated results of salinity field distribution of brine discharged by different water intake-outlet schemes of seawater desalination projects are shown in Figure 8. It can be seen that the impact of the brine discharge under different schemes on the increase of salinity in the nearby sea area is less than 0.6 units in most sea areas, although the impact of the increase of salinity in a small area near the discharge outlet is more than 2.3 units. With the continuous discharge of brine, the salinity of the sea area near the discharge outlet obviously increases. However, due to the water exchange between Aoshan Bay and the

where A = horizontal eddy viscosity.

MIKE21-AD/Transport module is based on the following non-linear vertically integrated 2-D equations of conservation of salinity which takes into account advection, dispersion, source and sink.

$$\frac{\partial h\bar{s}}{\partial t} + \frac{\partial h\bar{u}\bar{s}}{\partial x} + \frac{\partial h\bar{v}\bar{s}}{\partial y} = hF_s + h\bar{s}_s S \quad (6)$$

where, $\bar{s}$ = Depth average salinity, F_s = the horizontal diffusion term defined by

$$F_s = \left[\frac{\partial}{\partial x} \left(D_h \frac{\partial \bar{s}}{\partial x} \right) + \frac{\partial}{\partial y} \left(D_h \frac{\partial \bar{s}}{\partial y} \right) \right] \quad (7)$$

and D_h = horizontal diffusion coefficient.

In the 2D transport equations, the low order approximation uses simple first order upwinding, i.e., element average values in the upwinding direction are used as values at the boundaries. The higher order version approximates gradients to obtain second order accurate values at the boundaries. Values in the upwinding direction are used. To provide stability and minimize oscillatory effects, a TVD-MUSCL limiter is applied [5].

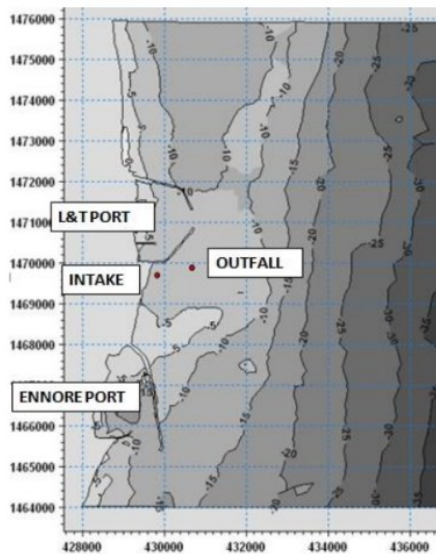
4. MODEL SIMULATION

4.1. Model Region

The region considered for the model is about 9 km X 12 km covering about 6 km on the either side of the proposed port and extending up to about -30 m contour as shown in Fig. 1. The western side of the model region consists the shoreline while the remaining sides are open to sea. The region is discretised by 11947 triangular elements and 6241 nodes (Fig.2). It is observed in the open sea at the site that the flow is unidirectional (either northerly or southerly). Hence to get the northerly flow, a tidal variation at southern boundary and velocities at northern boundary were considered. To get southern flow, the boundary conditions are reversed. At the seaward boundary, no flow was assumed.

4.2. Model Calibration

Before using the model for predictive purpose, it is calibrated. The Fig. 3A and 3B show calibration of the hydrodynamic model, particularly the observed and computed variation of



4.4. Model Operation

The ambient temperature of sea water is assumed to be 27°C. Ambient salinity of sea water is assumed to be 32 ppt. The rise in salinity of brine water to be discharged at the outfall is given to be 31 ppt above ambient.

For the salinity model, as the open boundaries have been taken sufficiently away from the outfall locations, zero rise in salinity is assumed at the open boundaries. It is observed that in the open sea at the site, the flow is unidirectional, either northerly or southerly generally parallel to the coastline. During the period February-September the flow is northward, while during the period October-January the flow is southward. The model is operated for different current conditions namely northward 0.25 m/s, southward 0.25 m/s and weak current i.e. 0.05 m/s towards. The direction of the weak current is considered to be northward or southward depending on the locations of intake and outfall to simulate the critical condition.

4.5. Model Operating Cases

The following cases of the intake and outfall were considered for the numerical model simulations.

Intake and outfall locations on south of south breakwater of L & T port as shown in Fig.1 are considered. Intake is taken at 550 m from shore. The outfall distance from shore is varied from 1000 m to 1300 m. Discharge of 1.25 m³/s (100 MLD) is considered at outfall while inflow at intake is taken twice that of outfall.. The following alternatives of outfall locations are considered.

1. With outfall at 1000 m from shore
2. With outfall at 1100 m from shore
3. With outfall at 1200 m from shore
4. With outfall at 1300 m from shore

4.6. Results and Discussion

The model is operated under northward, southward and weak current conditions in each of the above cases of outfall location. Typical results, namely, contour plots and time histories of rise in salinity are shown in Figs. 4A, 4B, 5A and 5B. The salinity rise at the intake is nearly zero in case of average northward current of 0.25 for all the outfall lengths. The salinity rise obtained at the intake for different outfall lengths and for the remaining two current conditions (namely, southward and weak current) are tabulated in Table 1A and 1B. It is seen that for the average northward current of 0.25 m/s the rise in salinity above ambient at the intake is well below 0.1 ppt. The salinity rise results from the tables are also plotted in the Fig. 6. From the tables and the Fig. 6 it is seen that it is also less than 0.1 ppt for average southward current of 0.25 m/s for the outfall distance greater than 1000 m and it is more than 0.1 ppt for outfall distance less than about 975 m.

Appendix B Complete Output Inventory

All figures and datasets generated by the UBDS solver for this case study:

B.1 Figures

- 01. [Near-field initial dilution] A_nearfield_RO_S1.png - Near-field trajectory, dilution and salinity - RO concentrate, 300 m³/hr (300 m³/hr per port).
- 02. [Near-field initial dilution] A_nearfield_RO_S2.png - Near-field trajectory, dilution and salinity - RO concentrate, 700 m³/hr (700 m³/hr per port).
- 03. [Near-field initial dilution] A_nearfield_RO_S3.png - Near-field trajectory, dilution and salinity - RO concentrate, 1200 m³/hr (600 m³/hr per port).
- 04. [Near-field initial dilution] A_nearfield_Sa_S1.png - Near-field trajectory, dilution and salinity - Saturated cavern brine, 300 m³/hr (300 m³/hr per port).
- 05. [Near-field initial dilution] A_nearfield_Sa_S2.png - Near-field trajectory, dilution and salinity - Saturated cavern brine, 700 m³/hr (700 m³/hr per port).
- 06. [Near-field initial dilution] A_nearfield_Sa_S3.png - Near-field trajectory, dilution and salinity - Saturated cavern brine, 1200 m³/hr (600 m³/hr per port).
- 07. [Near-field initial dilution] A_scenario_comparison.png - Seabed initial dilution by flow scenario and tidal state.
- 08. [Diffuser design optimisation] A2_angle_opt.png - Diffuser port-angle optimisation: dilution and seabed salinity vs discharge angle.
- 09. [Hydrodynamics, mesh & calibration] B_mesh_medium.png - Medium-field mesh & bathymetry (Bahia Azul)
- 10. [Hydrodynamics, mesh & calibration] B_mesh_far.png - Far-field (regional) domain & bathymetry
- 11. [Hydrodynamics, mesh & calibration] B_stations.png - Diffuser location & ADCP current-meter stations
- 12. [Hydrodynamics, mesh & calibration] B_adcp_A0.png - Modelled vs observed current speed & direction at station A0 (neap & spring).
- 13. [Hydrodynamics, mesh & calibration] B_adcp_A1.png - Modelled vs observed current speed & direction at station A1 (neap & spring).
- 14. [Hydrodynamics, mesh & calibration] B_adcp_A2.png - Modelled vs observed current speed & direction at station A2 (neap & spring).
- 15. [Hydrodynamics, mesh & calibration] B_adcp_A3.png - Modelled vs observed current speed & direction at station A3 (neap & spring).
- 16. [Hydrodynamics, mesh & calibration] B_adcp_A4.png - Modelled vs observed current speed & direction at station A4 (neap & spring).
- 17. [Hydrodynamics, mesh & calibration] B_tidal_level.png - Modelled vs observed tidal level (spring tide) at the calibration station.
- 18. [Hydrodynamics, mesh & calibration] B_flowfield_flood.png - Depth-averaged tidal flow field at flood (spring tide).
- 19. [Hydrodynamics, mesh & calibration] B_flowfield_ebb.png - Depth-averaged tidal flow field at ebb (spring tide).

- 20. [Medium-field dispersion] C_sigma_layers.png - Variation of the terrain-following sigma layers with water depth (9-layer scheme).
- 21. [Medium-field dispersion] C_layersens_4.png - Maximum seabed salinity - 4-layer model (full flow, neap)
- 22. [Medium-field dispersion] C_layersens_9.png - Maximum seabed salinity - 9-layer model (full flow, neap)
- 23. [Medium-field dispersion] C_layersens_14.png - Maximum seabed salinity - 14-layer model (full flow, neap)
- 24. [Medium-field dispersion] C_env_heights_panel.png - Maximum salinity vs height above the seabed - neap, full flow (dense plume is bottom-trapped and decays upward)
- 25. [Medium-field dispersion] C_snap_neap_high_water.png - Seabed salinity & current vectors at high water - neap
- 26. [Medium-field dispersion] C_vprof_neap_high_water.png - Vertical salinity profile at high water - neap
- 27. [Medium-field dispersion] C_snap_neap_mid_ebb.png - Seabed salinity & current vectors at mid ebb - neap
- 28. [Medium-field dispersion] C_vprof_neap_mid_ebb.png - Vertical salinity profile at mid ebb - neap
- 29. [Medium-field dispersion] C_snap_neap_low_water.png - Seabed salinity & current vectors at low water - neap
- 30. [Medium-field dispersion] C_vprof_neap_low_water.png - Vertical salinity profile at low water - neap
- 31. [Medium-field dispersion] C_snap_neap_mid_flood.png - Seabed salinity & current vectors at mid flood - neap
- 32. [Medium-field dispersion] C_vprof_neap_mid_flood.png - Vertical salinity profile at mid flood - neap
- 33. [Medium-field dispersion] C_snap_spring_high_water.png - Seabed salinity & current vectors at high water - spring
- 34. [Medium-field dispersion] C_snap_spring_mid_ebb.png - Seabed salinity & current vectors at mid ebb - spring
- 35. [Medium-field dispersion] C_snap_spring_low_water.png - Seabed salinity & current vectors at low water - spring
- 36. [Medium-field dispersion] C_snap_spring_mid_flood.png - Seabed salinity & current vectors at mid flood - spring
- 37. [Medium-field dispersion] C_env_springneap.png - Maximum seabed salinity over a full spring-neap cycle (full flow)
- 38. [Far-field dispersion] D_far_neap_scenarios.png - Far-field maximum seabed salinity by discharge scenario - neap tide (RO brine)
- 39. [Far-field dispersion] D_far_spring_S3.png - Far-field maximum seabed salinity - spring, full flow (RO brine)
- 40. [Far-field dispersion] D_far_springneap.png - Far-field maximum seabed salinity - spring-neap envelope (full flow)
- 41. [Far-field dispersion] D_far_cavern.png - Far-field maximum seabed salinity - phase-2 saturated cavern brine (worst case)

- 42. [Design comparison & optimisation] E_salinity_transect.png - Maximum seabed salinity along a transect through the diffuser.
- 43. [Design comparison & optimisation] E_current_curve.png - Maximum current speed along the outfall corridor (spring tide).
- 44. [Design comparison & optimisation] E_constraints.png - Distribution of marine development and environmental constraints relative to the mixing zone.
- 45. [Design comparison & optimisation] E_optimised_layout.png - Recommended optimised diffuser/intake layout (inclined two-port diffuser).
- 46. [Outfall siting & recirculation] H_intake_rise_curve.png - Variation of salinity rise at the intake with outfall distance from shore (recirculation criterion).
- 47. [Outfall siting & recirculation] H_rise_contour_near.png - Salinity-rise contours, outfall 600 m from shore, weak southward current.
- 48. [Outfall siting & recirculation] H_rise_contour_far.png - Salinity-rise contours, outfall 1200 m from shore, weak southward current.
- 49. [Outfall siting & recirculation] H_timehist_near.png - Time history of salinity rise at intake and outfall, outfall 600 m from shore.
- 50. [Outfall siting & recirculation] H_timehist_far.png - Time history of salinity rise at intake and outfall, outfall 1200 m from shore.
- 51. [Inputs, parameters & water quality] F_impurities.png - Predicted receiving-water trace-impurity concentrations vs EQS thresholds.
- 52. [Data plots] G_dilution_vs_velocity.png - Seabed dilution as a function of tidal velocity for each brine type and scenario.
- 53. [Data plots] G_area_vs_flow.png - Area enclosed by salinity-increment contours vs discharge flow.
- 54. [Data plots] G_skill.png - Index-of-agreement between modelled and observed currents by station (spring).
- 55. [Reference output figures (source studies)] appendix_b_fig_p8.png - Reference Study 1 (salt-cavern brine outfall): model mesh, bathymetry and current-meter comparison.
- 56. [Reference output figures (source studies)] appendix_b_fig_p25.png - Reference Study 1: near-field brine-plume trajectory and initial-dilution prediction.
- 57. [Reference output figures (source studies)] appendix_b_fig_p33.png - Reference Study 1: sigma-layer scheme and layer-resolution sensitivity.
- 58. [Reference output figures (source studies)] appendix_b_fig_p39.png - Reference Study 1: maximum seabed-salinity envelope and salinity/current snapshots.
- 59. [Reference output figures (source studies)] appendix_b_fig_p55.png - Reference Study 1: far-field maximum seabed-salinity envelopes.
- 60. [Reference output figures (source studies)] water_15_02402_fig_p4.png - Reference Study 2 (Laoshan Bay desalination): computational mesh and bathymetry of the bay.
- 61. [Reference output figures (source studies)] water_15_02402_fig_p6.png - Reference Study 2: tidal flow-field simulation results.
- 62. [Reference output figures (source studies)] water_15_02402_fig_p7.png - Reference Study 2: salinity-field results and salinity profile for the layout plans.
- 63. [Reference output figures (source studies)] ijciat_fig_p4.png - Reference Study 3 (Chennai desalination): bathymetry and flexible computational mesh.

- 64. [Reference output figures (source studies)] ijciet_fig_p5.png - Reference Study 3: hydrodynamic calibration (observed vs computed tidal level and current).
- 65. [Reference output figures (source studies)] ijciet_fig_p7.png - Reference Study 3: contours of salinity rise and time history at intake/outfall.
- 66. [Validation] V_jet.png - Momentum-jet dilution: UBDS vs the canonical self-similar law (Fischer et al. 1979).
- 67. [Validation] V_plume.png - Pure-plume dilution: UBDS vs the 5/3 power law (Morton, Taylor & Turner 1956).
- 68. [Validation] V_dense_jet.png - Inclined 60 deg dense-jet dimensionless rise and dilution vs published experimental bands.
- 69. [Animations] ANIM_brine_dispersion_filmstrip.png - Brine dispersion over a tidal cycle (full flow, spring) (filmstrip of the animation).

B.2 Animations

- 01. ANIM_brine_dispersion_spring.gif - Brine dispersion over a tidal cycle (full flow, spring)
- 02. ANIM_brine_dispersion_neap.gif - Brine dispersion over a tidal cycle (full flow, neap)
- 03. ANIM_current_field_spring.gif - Tidal current field over a cycle (spring)

B.3 Datasets (CSV)

- 01. A_nearfield_initial_dilution.csv - Near-field initial-dilution results for all brine types, flow scenarios and tidal states.
- 02. A2_port_angle_optimisation.csv - Effect of diffuser port discharge angle on near-field dilution (300 m³/hr, neap).
- 03. B_current_validation_skill.csv - Hydrodynamic calibration skill (RMSE, Willmott index, Nash-Sutcliffe) at ADCP stations.
- 04. B_tidal_level.csv - Tidal-level validation time series.
- 05. D_influence_area_diffusion.csv - Influence areas of salinity increments and diffusion distances per flow scenario (far field).
- 06. E_salinity_transect.csv - Seabed salinity transect data.
- 07. H_intake_recirculation.csv - Salinity rise above ambient at the intake vs outfall distance from shore, by current condition (0.1 ppt recirculation criterion).
- 08. H_table1A_southward.csv - Rise in salinity at intake above ambient - southward 0.25 m/s.
- 09. H_table1B_weak.csv - Rise in salinity at intake above ambient - weak southward 0.05 m/s.
- 10. H_timehist_near.csv - Time-history data, outfall 600 m from shore.
- 11. H_timehist_far.csv - Time-history data, outfall 1200 m from shore.
- 12. F_layer_scheme.csv - Sigma-layer schemes: percentage of water-column depth per layer (bottom layer last).
- 13. F_model_parameters.csv - Hydrodynamic and transport model parameters.
- 14. F_impurities_eqs.csv - Trace-impurity concentrations in the brine vs EQS after near-field dilution.
- 15. V_eos_validation.csv - Equation-of-state validation against reference seawater/brine densities.
- 16. V_jet.csv - Round-jet dilution validation data.

- 17. V_dense_jet.csv - Inclined (60 deg) dense-jet dimensionless geometry/dilution vs published ranges.